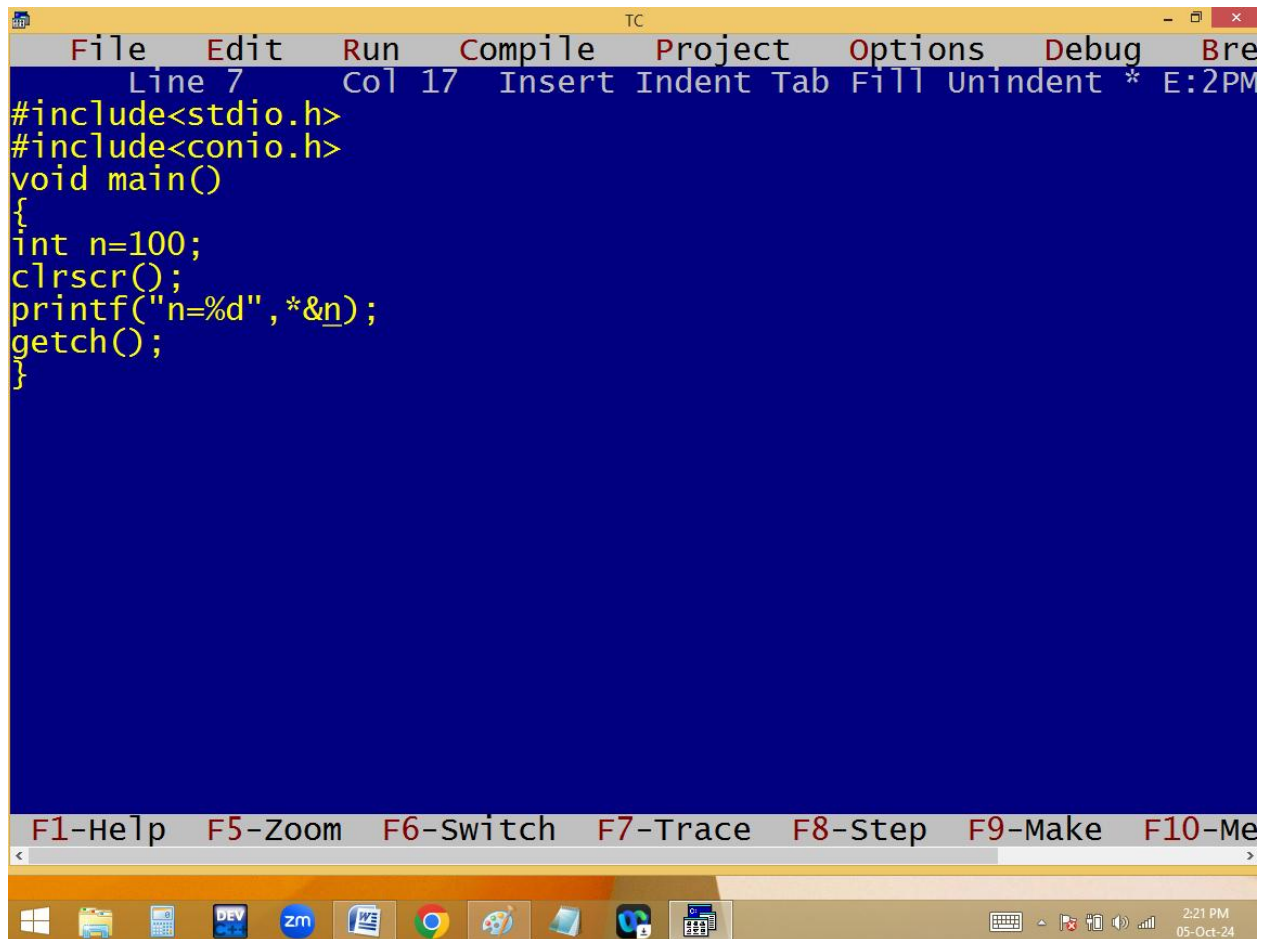


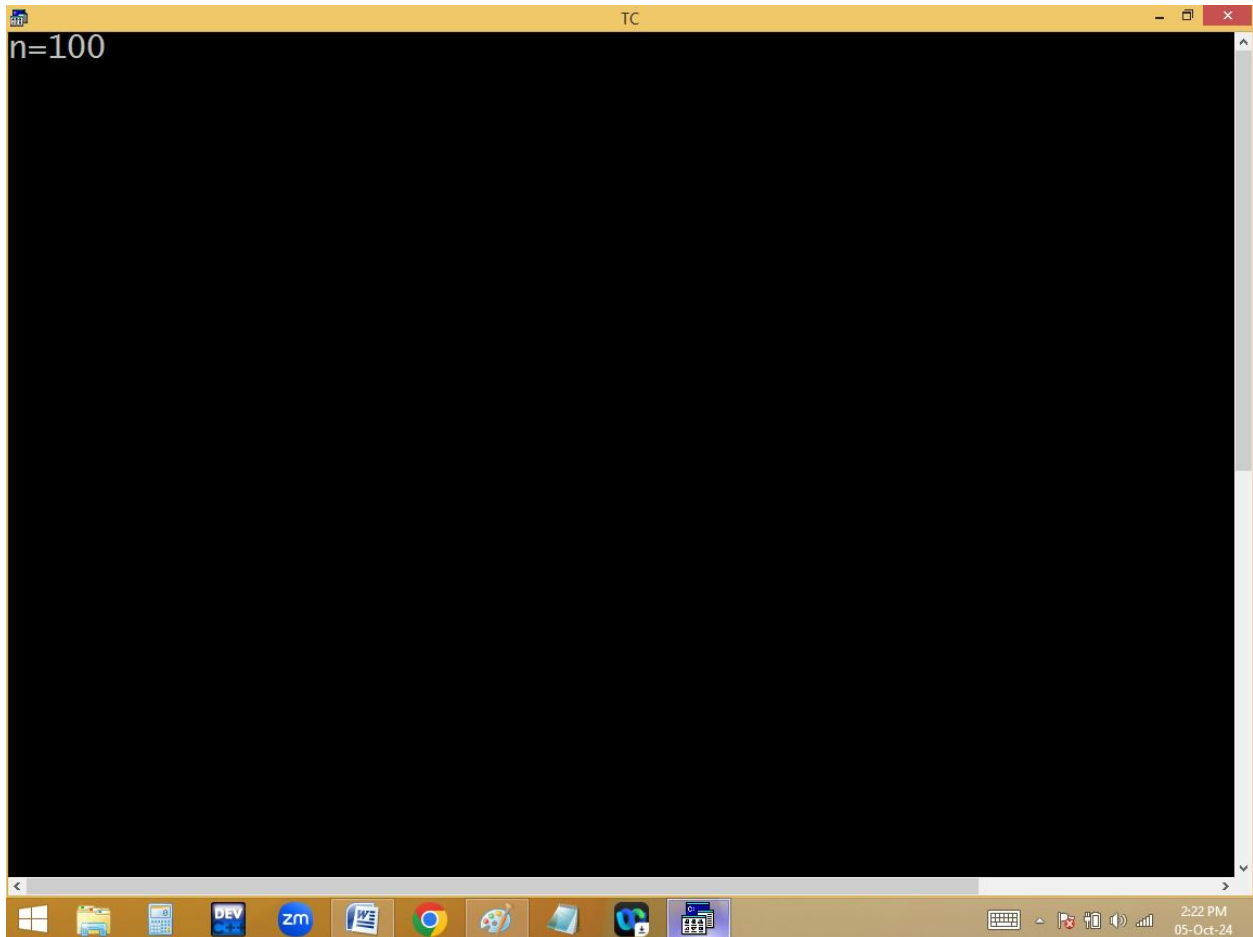
Finding a normal variable value using pointer mechanism:



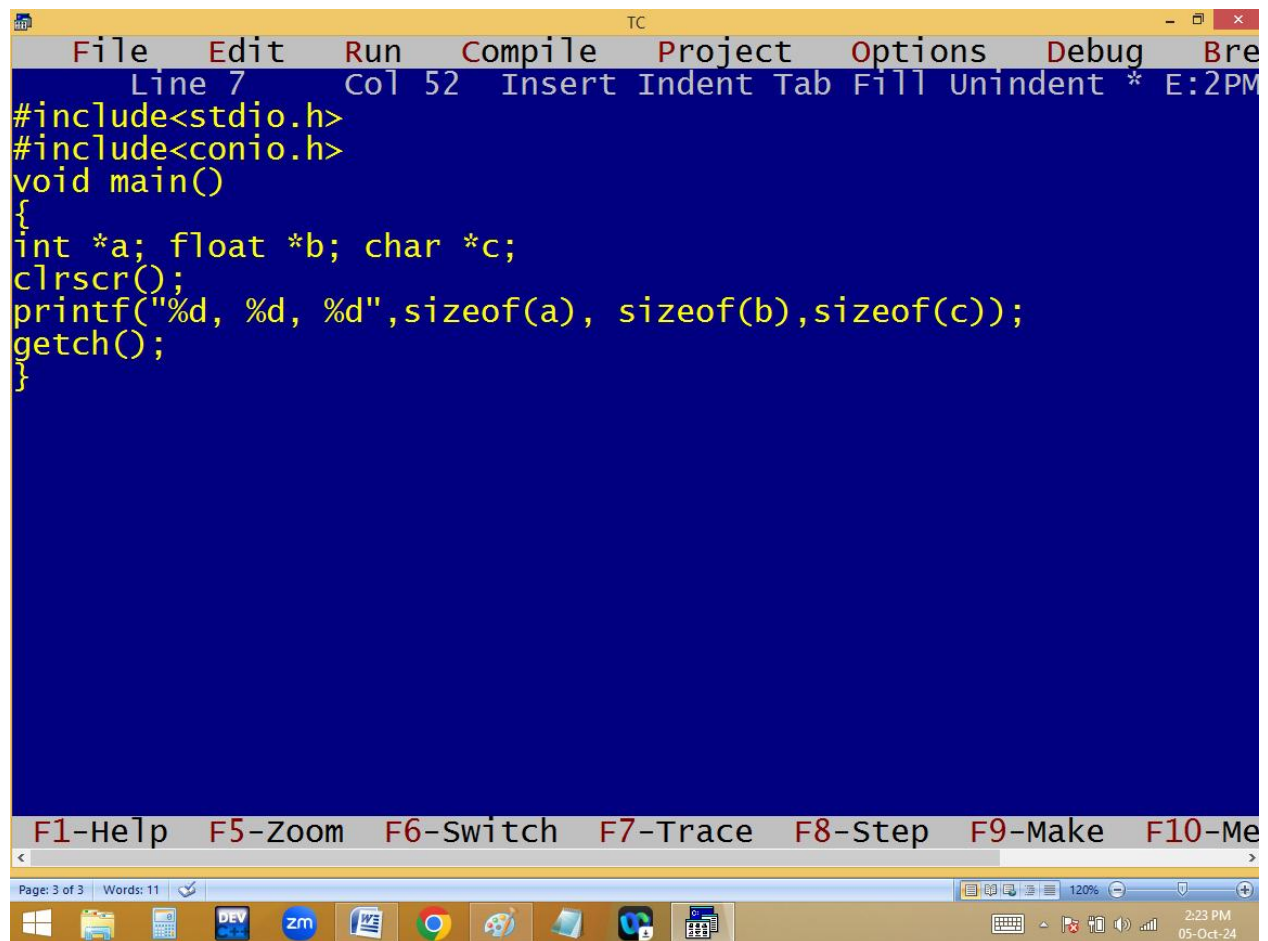
The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", "Debug", and "Breakpoint". The status bar at the top indicates "Line 7", "Col 17", and "Insert Indent Tab Fill Unindent * E:2PM". The main editing area has a dark blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n=100;
clrscr();
printf("n=%d", *&n);
getch();
}
```

Below the code editor, there is a toolbar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Me". The Windows taskbar is visible at the bottom, showing icons for the Start menu, File Explorer, Calculator, DEV C++, Zoom, Word, Chrome, Paint, and other applications. The system clock in the bottom right corner shows "2:21 PM" and "05-Oct-24".



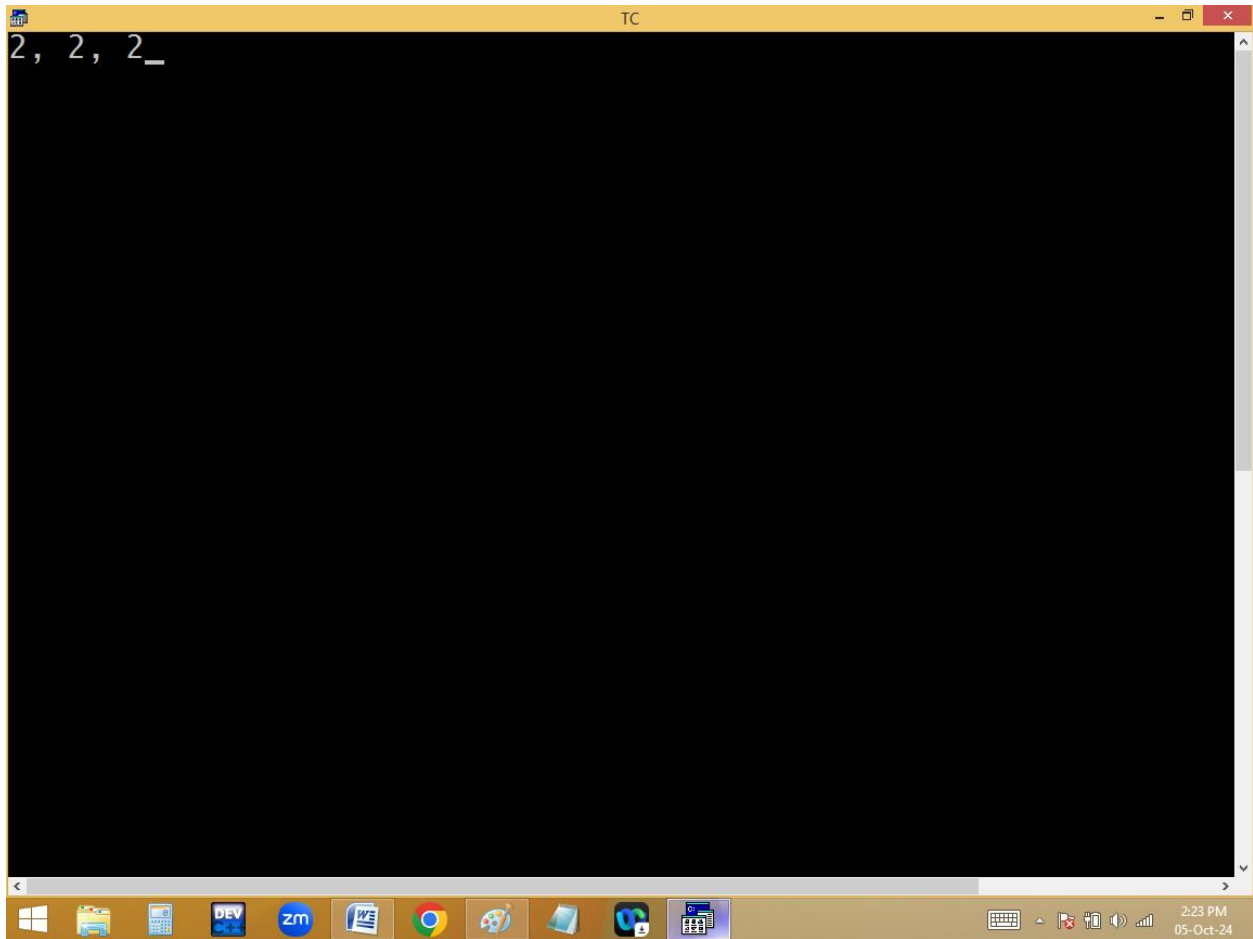
Finding pointer size: Any type of pointer it stores only the address and address is always unsigned int. due to this any type of pointer it takes **2 / 4 / 8** bytes in **16 / 32 / 64 bit compilers**.



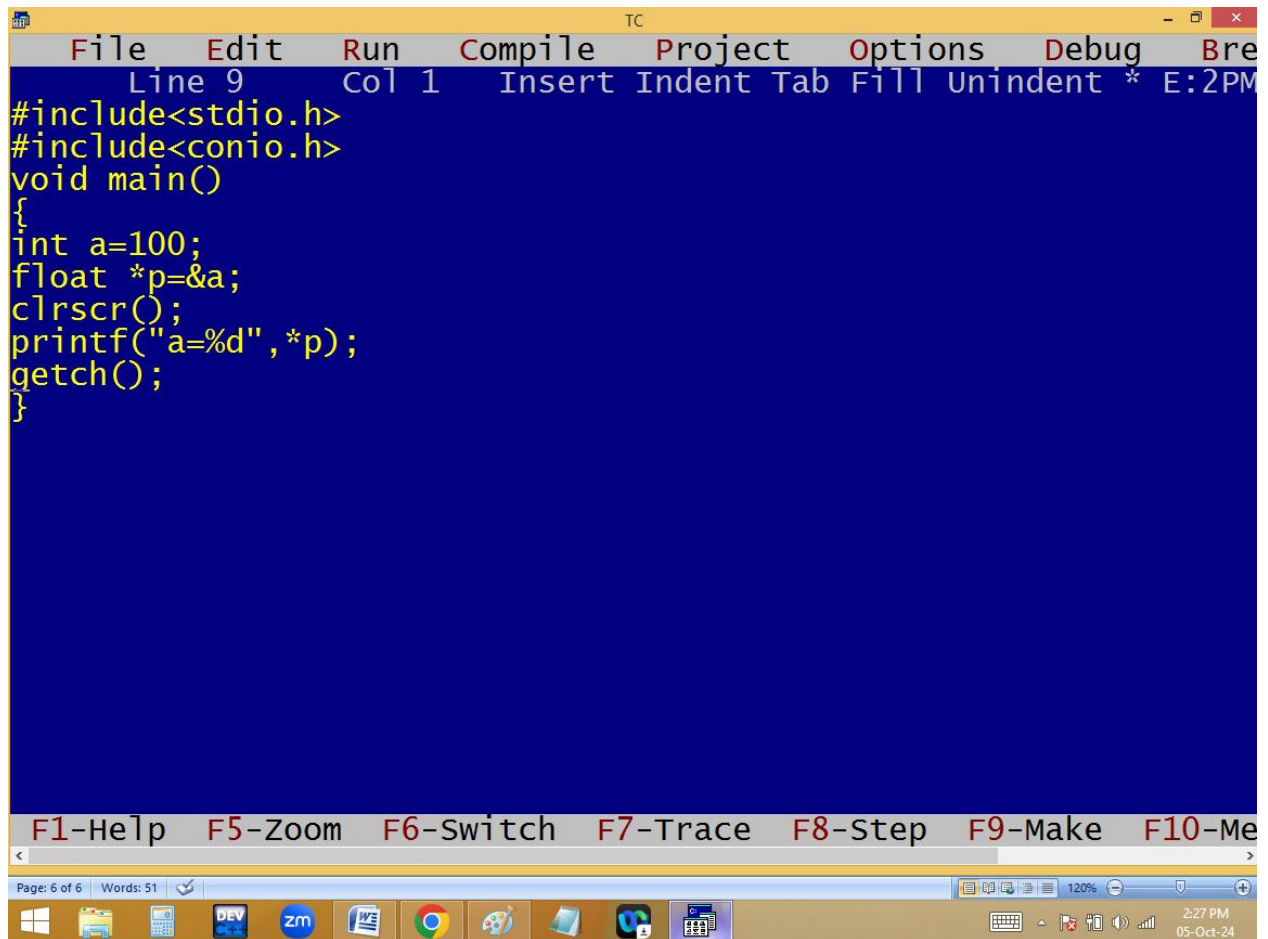
```
TC
File Edit Run Compile Project Options Debug Bre
Line 7 Col 52 Insert Indent Tab Fill Unindent * E:2PM
#include<stdio.h>
#include<conio.h>
void main()
{
int *a; float *b; char *c;
clrscr();
printf("%d, %d, %d",sizeof(a), sizeof(b),sizeof(c));
getch();
}
```

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Me

Page: 3 of 3 Words: 11 120% 2:23 PM 05-Oct-24



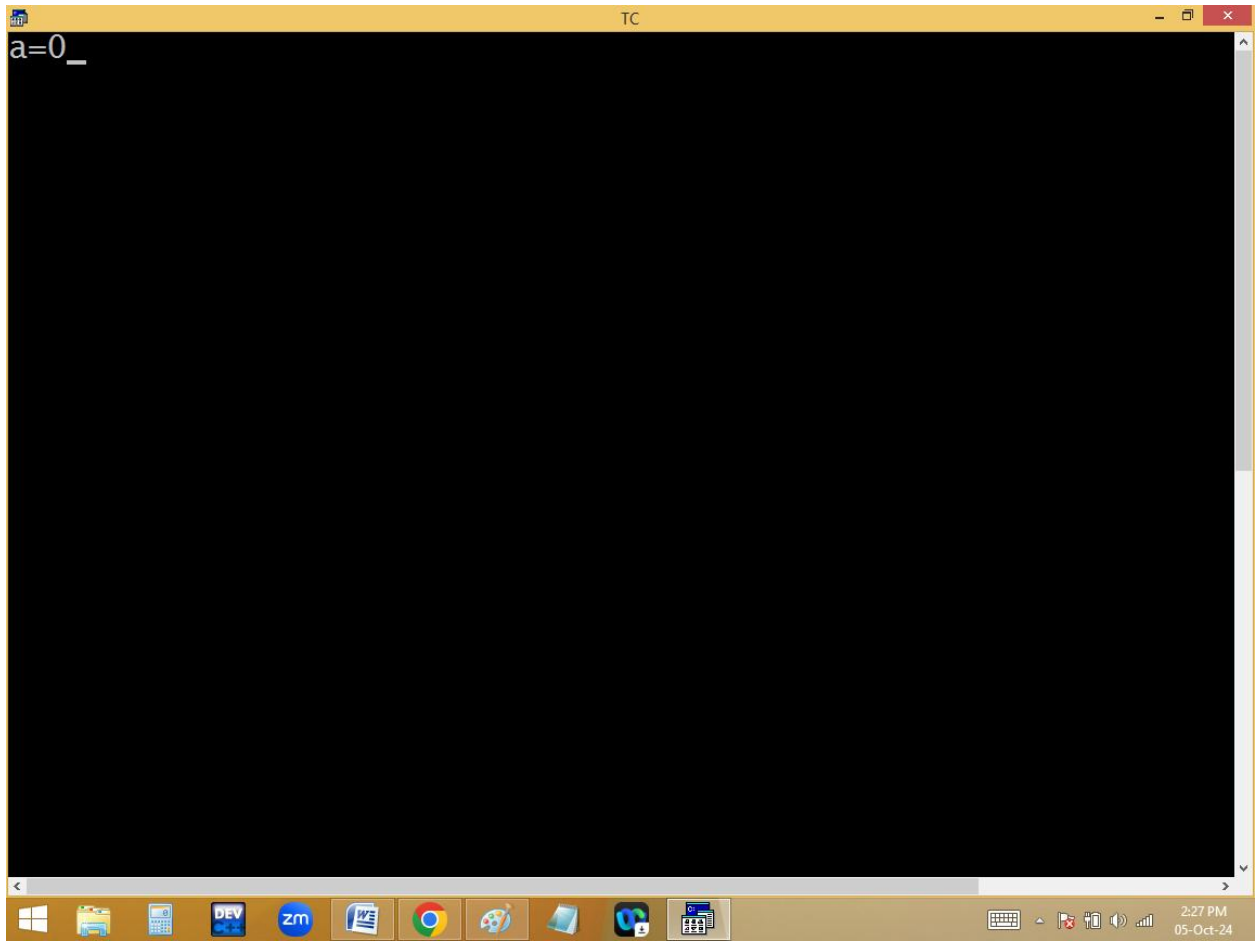
Pointer compatibility: pointer can store only the same type of variable address. When different type is given, it gives garbage or runtime error.



The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break. The status bar at the top shows "Line 9", "Col 1", and "E:2PM". The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a=100;
    float *p=&a;
    clrscr();
    printf("a=%d",*p);
    getch();
}
```

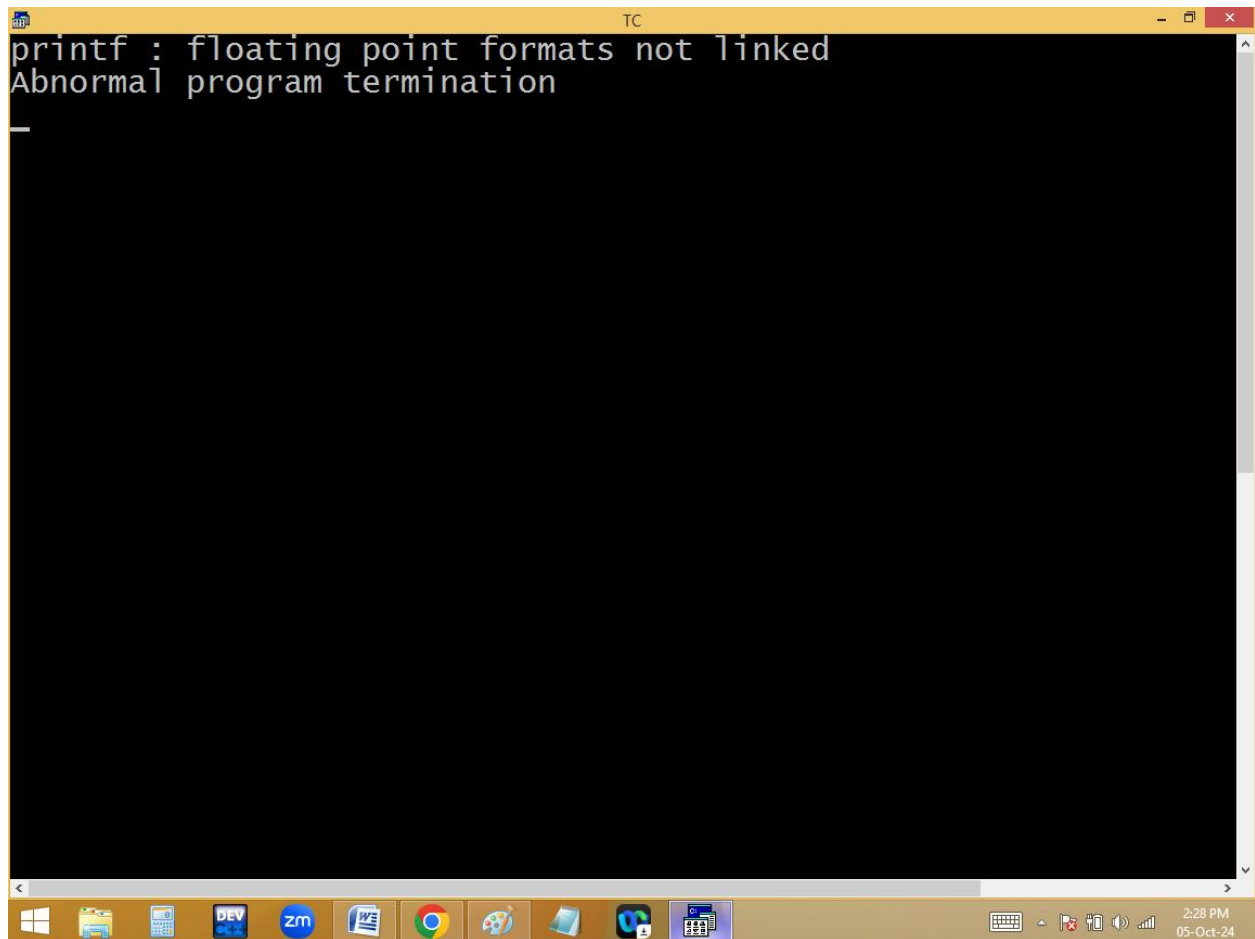
Below the code, there is a function key bar with shortcuts: F1-Help, F5-Zoom, F6-Switch, F7-Trace, F8-Step, F9-Make, and F10-Me. The bottom status bar shows "Page: 6 of 6", "Words: 51", and a zoom level of "120%". The Windows taskbar at the bottom includes icons for Windows, File Explorer, Calculator, DEV C++, Zoom, Word, Chrome, Paint, and other applications. The system clock shows "2:27 PM" and "05-Oct-24".



The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break. The status bar at the top indicates "Line 8", "Col 13", and "E:2PM". The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
float a=100;
int *p=&a;
clrscr();
printf("a=%f",*p);
getch();
}
```

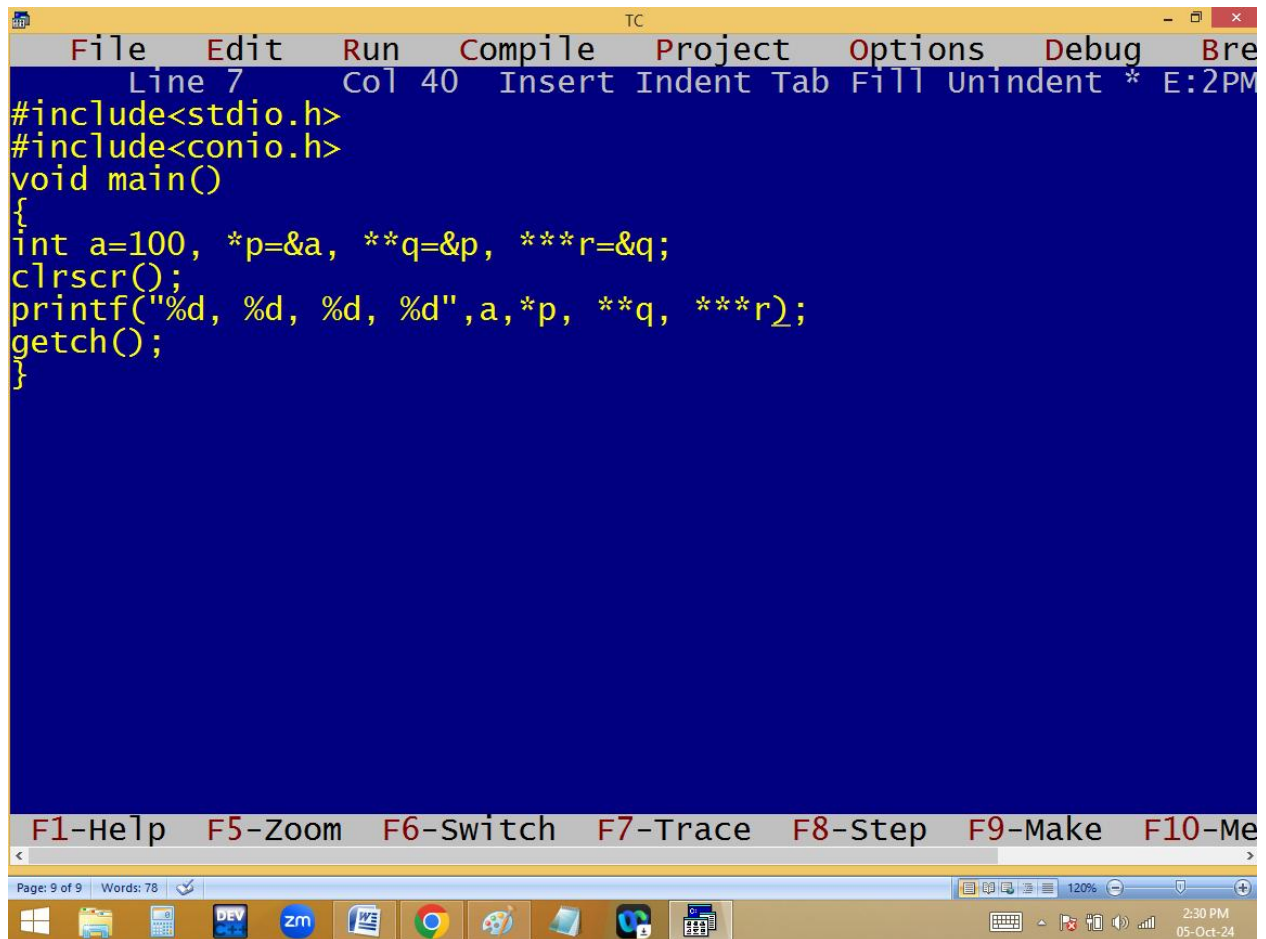
Below the code area, there is a toolbar with function key shortcuts: F1-Help, F5-Zoom, F6-Switch, F7-Trace, F8-Step, F9-Make, and F10-Me. The bottom status bar shows "Page: 6 of 7", "Words: 51", and a zoom level of "120%". The Windows taskbar at the very bottom displays various application icons including Windows Explorer, Calculator, DEV C++, Zoom, Word, Chrome, and others. The system clock in the bottom right corner shows "2:28 PM" and "05-Oct-24".



The image shows a screenshot of a Turbo C++ (TC) compiler window. The window has a yellow title bar with the text "TC" and standard minimize, maximize, and close buttons. The main area is a black console window with white text. The text displayed is: "printf : floating point formats not linked" followed by "Abnormal program termination" on the next line. Below the console window is a Windows taskbar with various application icons including Windows Explorer, Calculator, DEV C++, Zm, Word, Chrome, and others. The system tray on the right shows the time as 2:28 PM and the date as 05-Oct-24.

```
printf : floating point formats not linked
Abnormal program termination
```

Double pointer / pointer to pointer: The pointer which stores the address of another pointer is called double pointer. They are used to manage dynamic multi dimensional array.

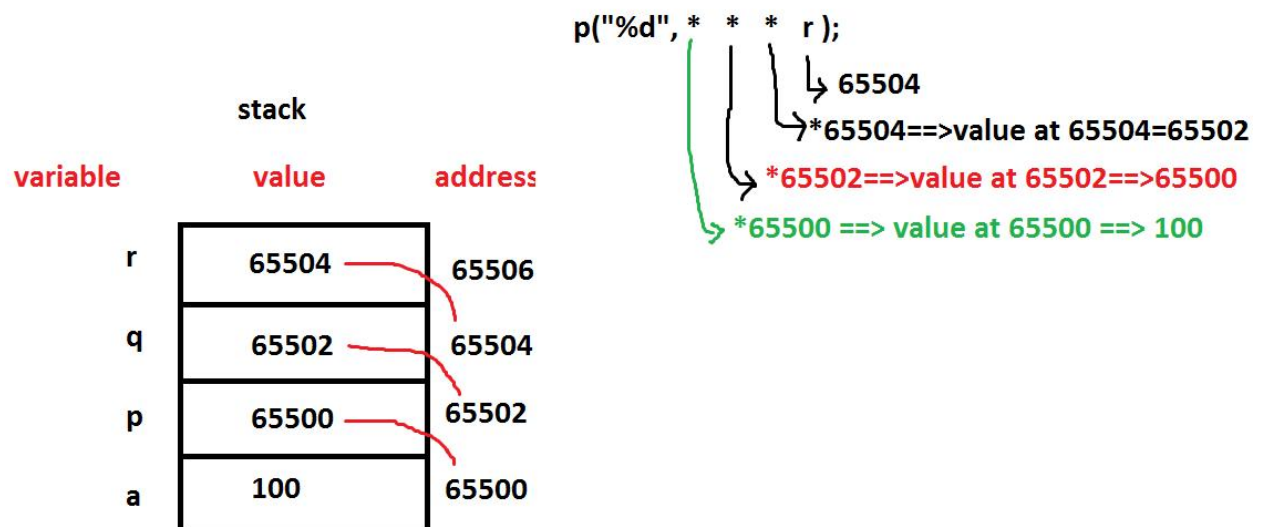
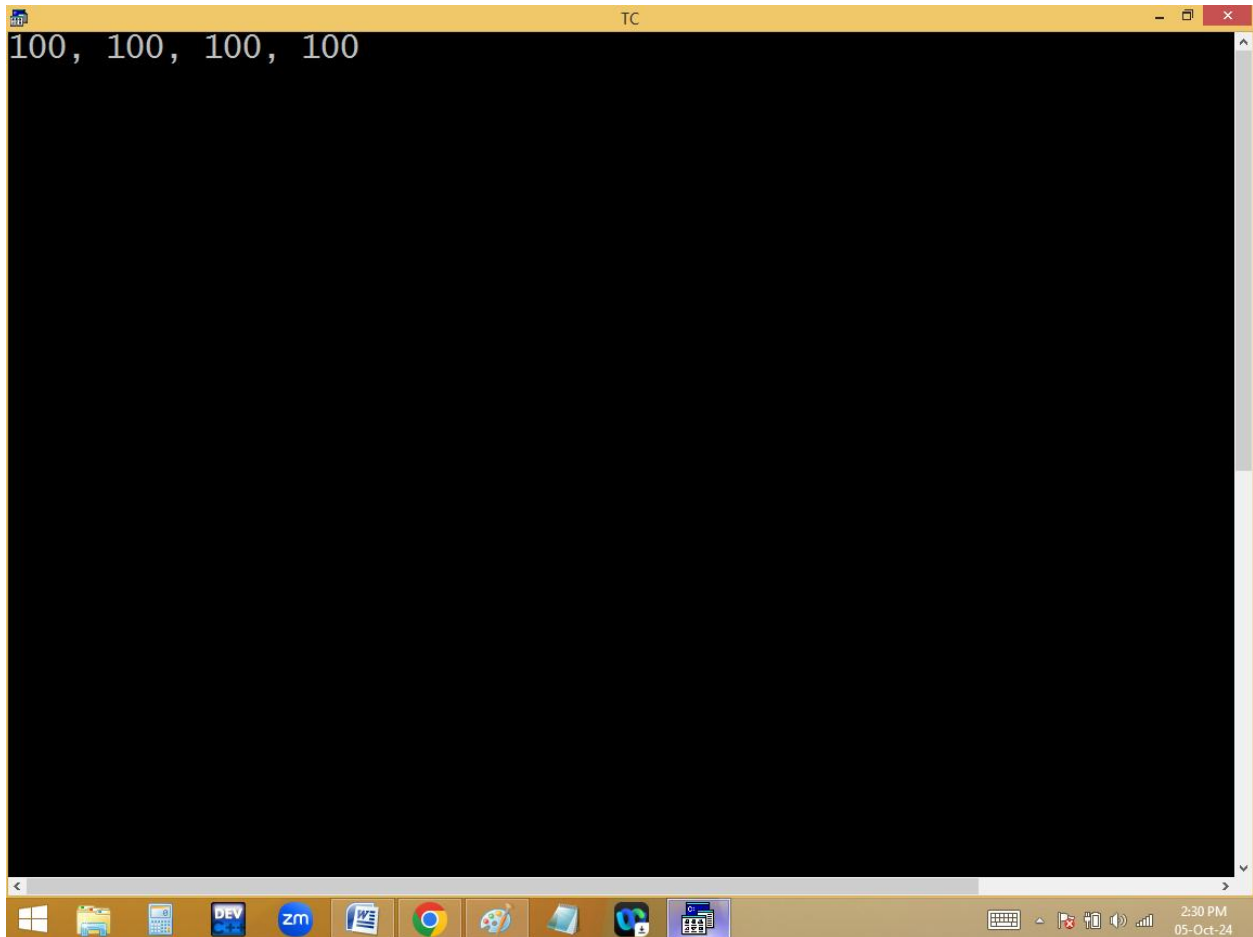


The image shows a screenshot of the Turbo C++ (TC) IDE. The main window has a dark blue background with yellow text. The menu bar at the top includes File, Edit, Run, Compile, Project, Options, Debug, and Break. The status bar at the top shows 'Line 7', 'Col 40', and 'E:2PM'. The code in the editor is as follows:

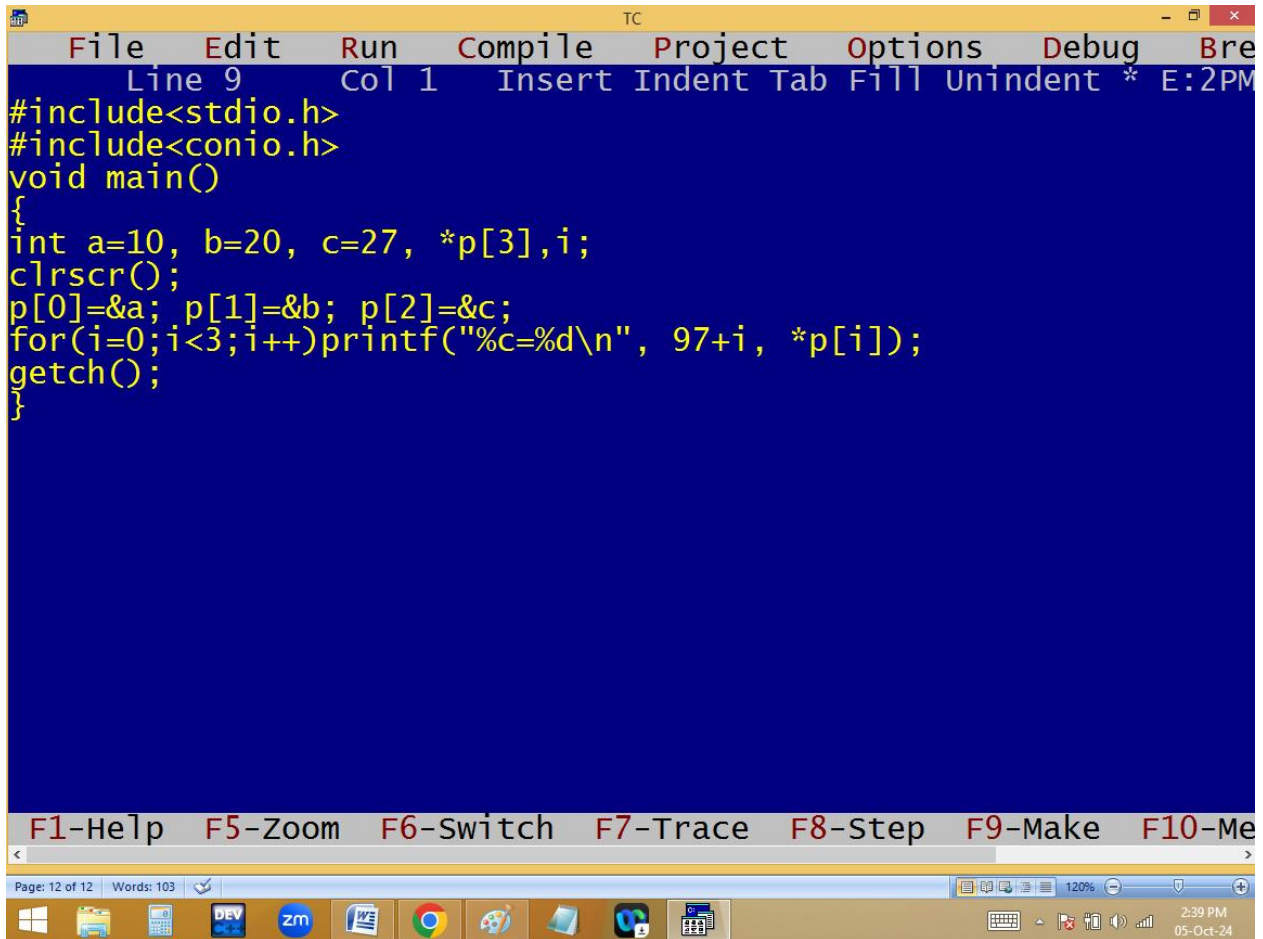
```
#include<stdio.h>
#include<conio.h>
void main()
{
int a=100, *p=&a, **q=&p, ***r=&q;
clrscr();
printf("%d, %d, %d, %d",a,*p, **q, ***r);
getch();
}
```

Below the code editor, there is a toolbar with icons for various functions. At the bottom, a status bar shows 'Page: 9 of 9' and 'Words: 78'. The Windows taskbar at the very bottom displays several application icons, including Windows Explorer, Calculator, DEV C++, Zoom, Word, Chrome, and others. The system clock in the bottom right corner shows '2:30 PM' and '05-Oct-24'.

File Edit Run Compile Project Options Debug Break
Line 7 Col 40 Insert Indent Tab Fill Unindent * E:2PM
#include<stdio.h>
#include<conio.h>
void main()
{
int a=100, *p=&a, **q=&p, ***r=&q;
clrscr();
printf("%d, %d, %d, %d",a,*p, **q, ***r);
getch();
}
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Me
Page: 9 of 9 Words: 78 120% 2:30 PM 05-Oct-24

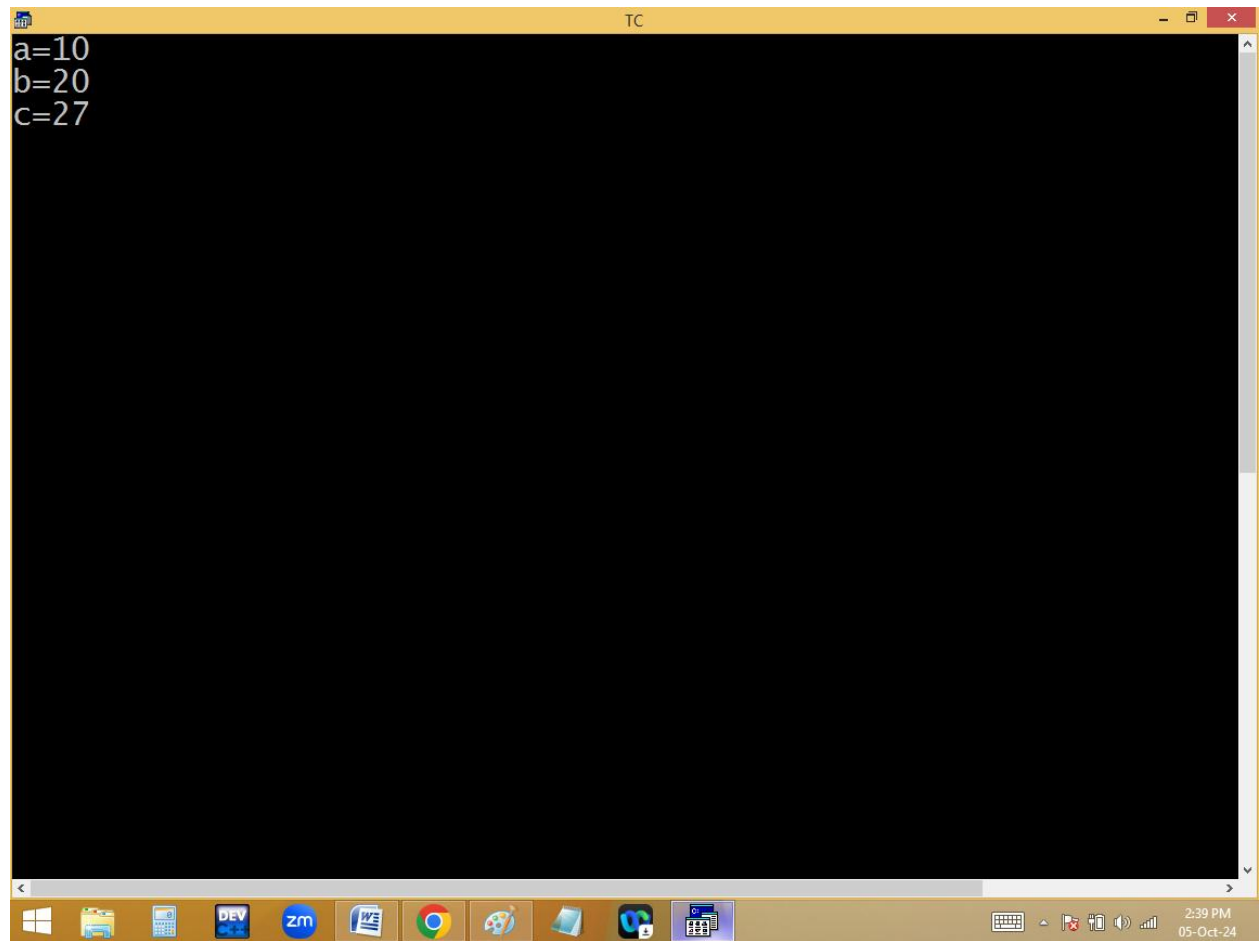


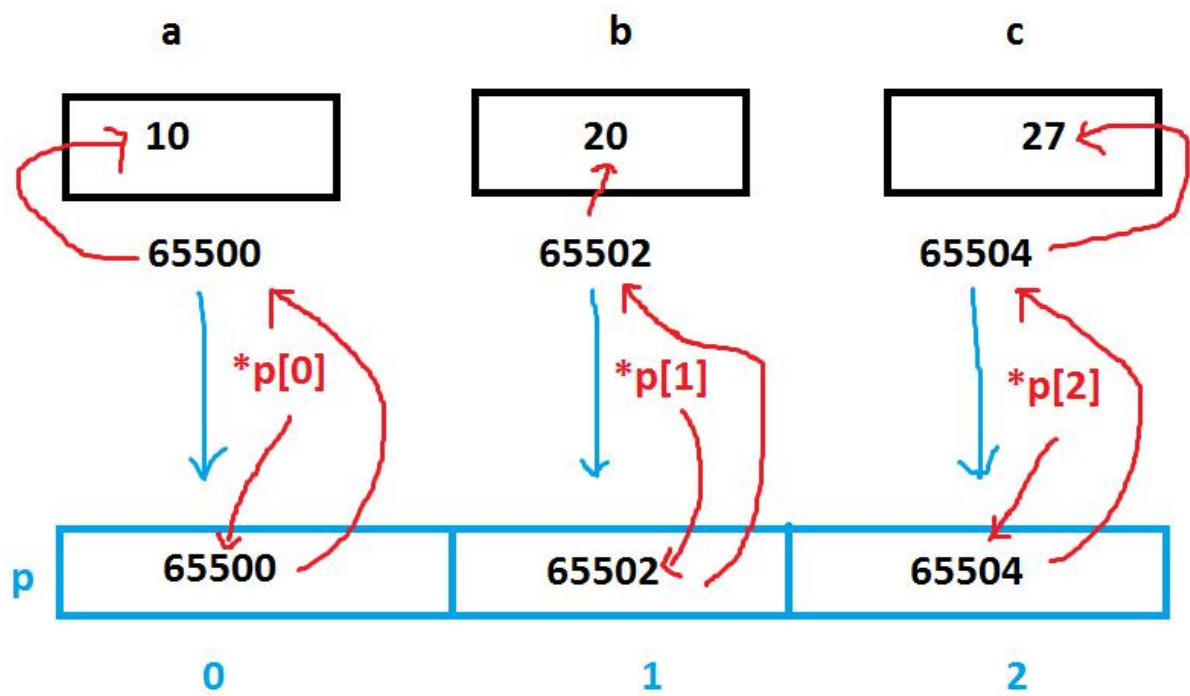
Array of pointer: Like normal variables we can define pointer with array. Due to this one pointer variable can store multiple address. It is used to handle dynamic multi dimensional array.

A screenshot of a Turbo C++ IDE window. The title bar says "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break. The status bar shows "Line 9", "Col 1", and "E:2PM". The code is as follows:

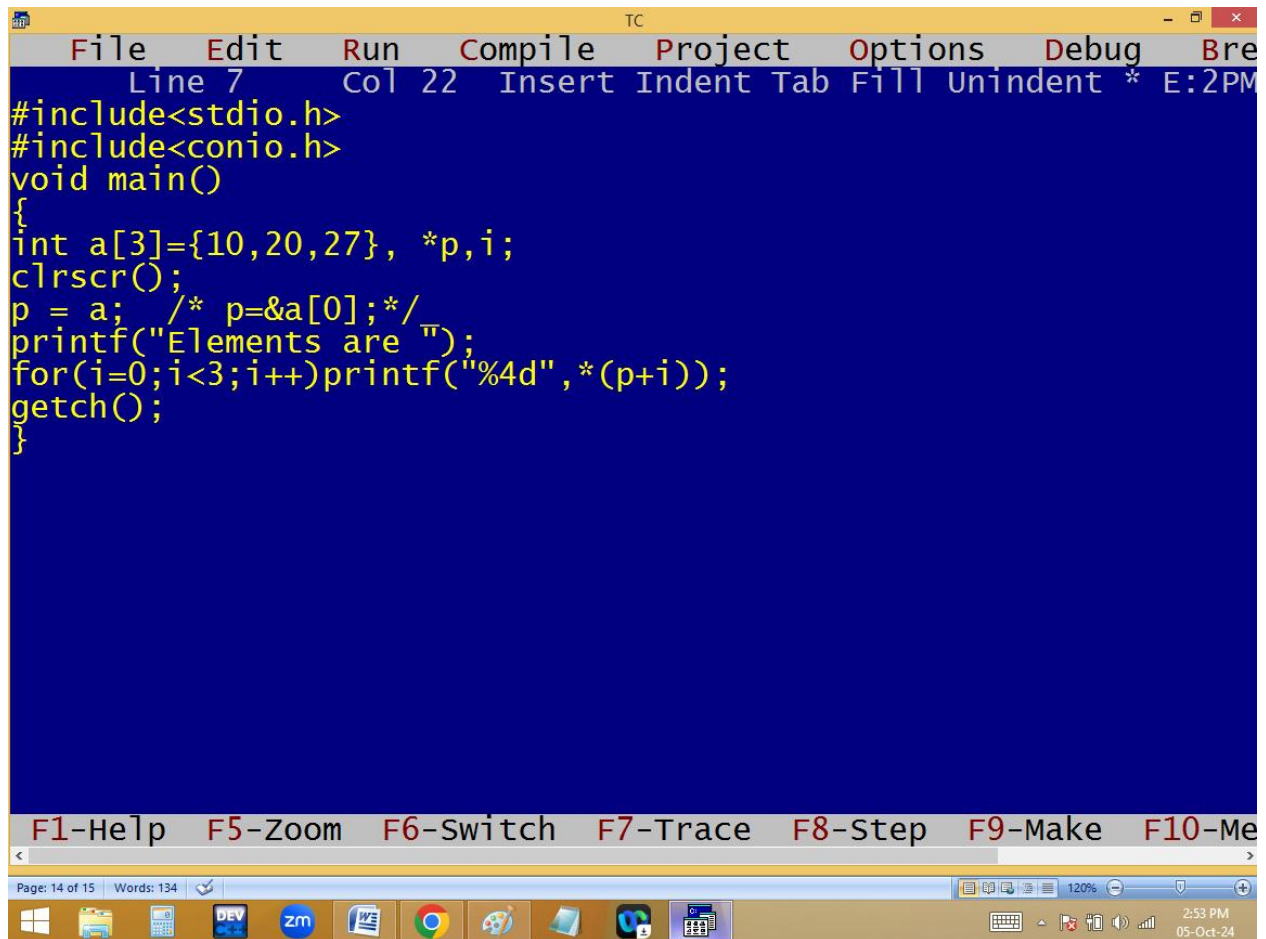
```
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10, b=20, c=27, *p[3],i;
clrscr();
p[0]=&a; p[1]=&b; p[2]=&c;
for(i=0;i<3;i++)printf("%c=%d\n", 97+i, *p[i]);
getch();
}
```

The bottom of the window shows a toolbar with icons for various functions and a status bar with "Page: 12 of 12", "Words: 103", and a taskbar with various application icons. The system clock in the bottom right corner shows "2:39 PM" and "05-Oct-24".





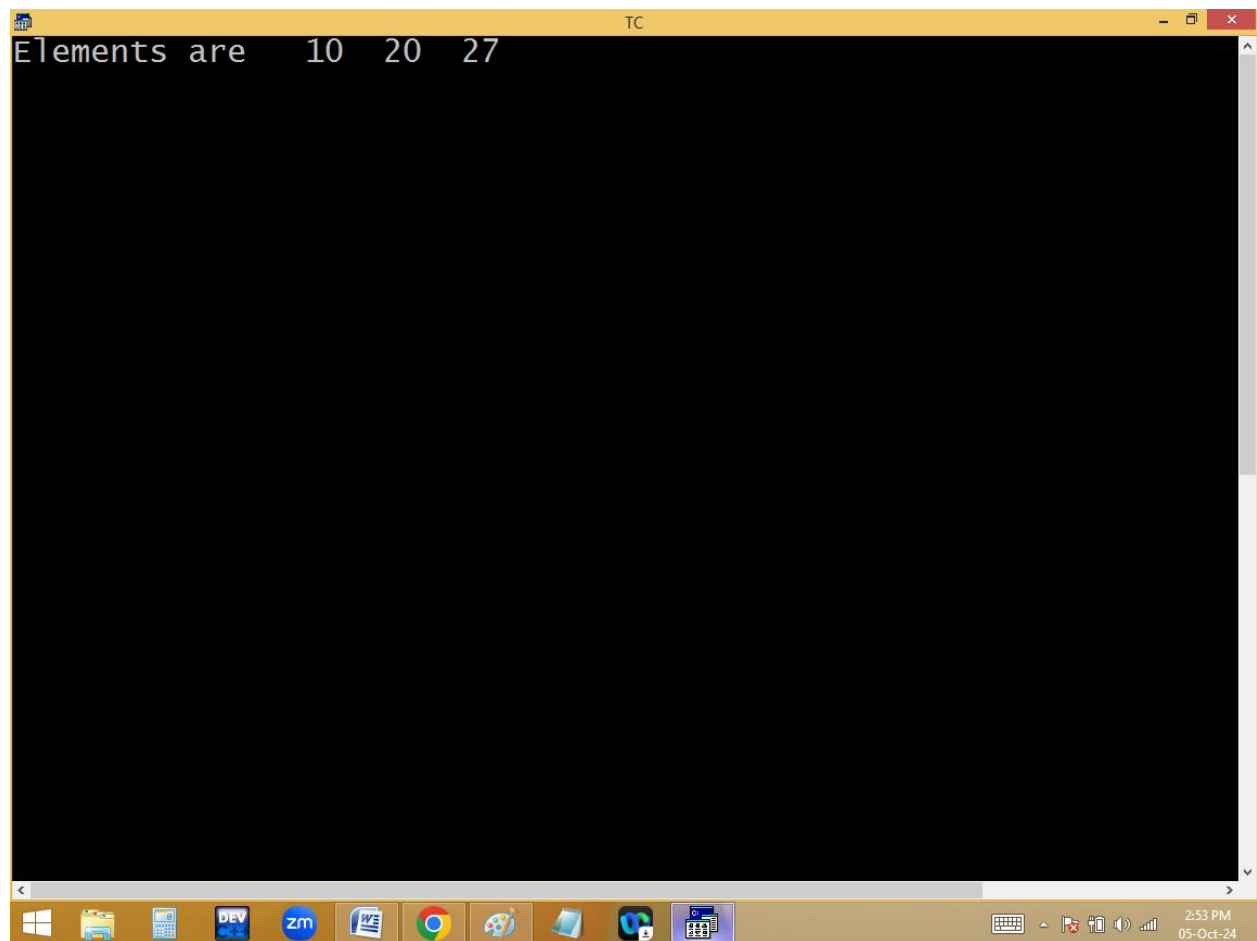
Pointer to array:

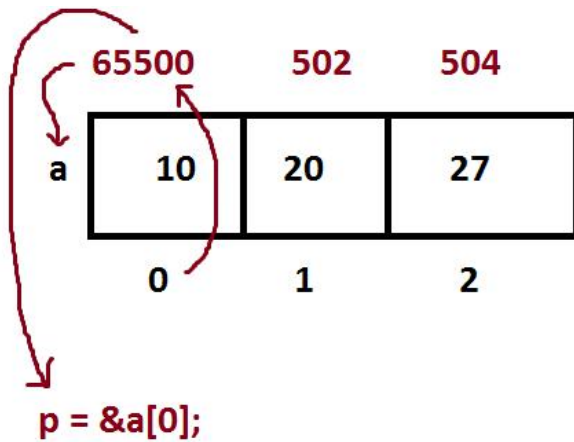


The image shows a screenshot of a Turbo C++ (TC) IDE. The main window has a dark blue background with yellow text for the code. The menu bar at the top includes File, Edit, Run, Compile, Project, Options, Debug, and Break. The status bar at the top shows 'Line 7', 'Col 22', and 'Insert Indent Tab Fill Unindent * E:2PM'. The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a[3]={10,20,27}, *p,i;
    clrscr();
    p = a; /* p=&a[0]; */
    printf("Elements are ");
    for(i=0;i<3;i++)printf("%4d",*(p+i));
    getch();
}
```

Below the code editor, there is a toolbar with icons for various functions. At the bottom of the IDE window, a status bar displays 'Page: 14 of 15' and 'Words: 134'. The Windows taskbar is visible at the very bottom, showing icons for Windows, File Explorer, Calculator, DEV C++, Zoom, Word, Chrome, Paint, and other applications. The system clock in the bottom right corner indicates the time is 2:53 PM on 05-Oct-24.





```
for( i=0; i<3; i++ ) printf("%d", *( p + i ));
```

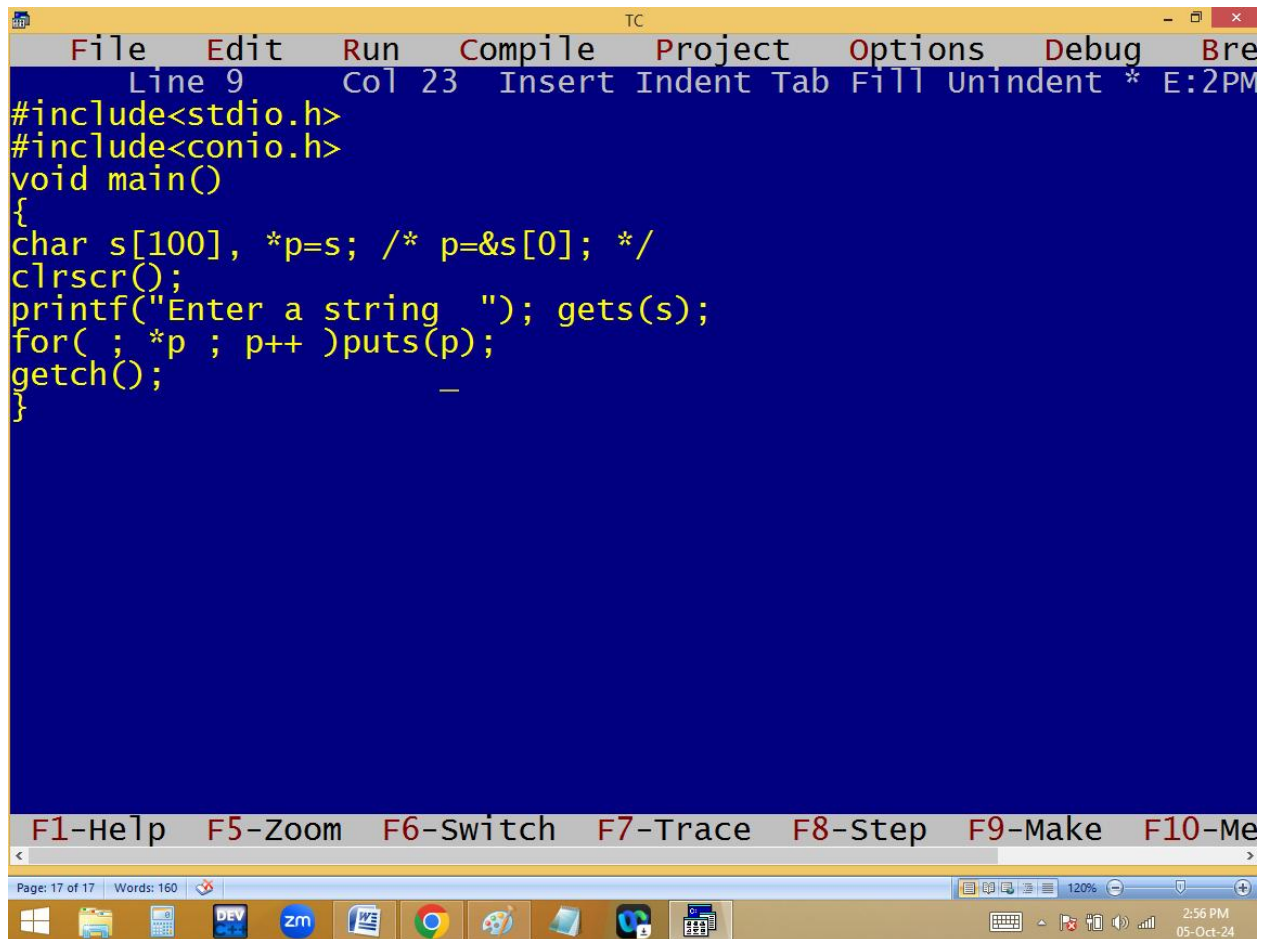
|

```
*65500+0*2=*65500==>value at 65500 ==> 10
*65500+1*2=*65502==>value at 65502 ==> 20
*65500+2*2=*65504==>value at 65504 ==> 27
```

We can access array elements in following ways:

`a[i] / i[a] / p[i] / i[p] / *(p+i) / *(a+i) / *(i+a) / *(i+p)`

pointer to string:

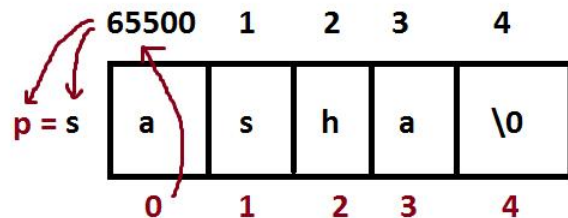


The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", "Debug", and "Breakpoint". The status bar at the top indicates "Line 9", "Col 23", and "Insert Indent Tab Fill Unindent * E:2PM". The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
char s[100], *p=s; /* p=&s[0]; */
clrscr();
printf("Enter a string "); gets(s);
for( ; *p ; p++ )puts(p);
getch();
}
```

Below the code, there is a toolbar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Me". At the bottom, a status bar shows "Page: 17 of 17" and "Words: 160". The Windows taskbar is visible at the very bottom, showing icons for Windows, File Explorer, Calculator, DEV C++, Zoom, Word, Chrome, and other applications. The system clock in the bottom right corner shows "2:56 PM" and "05-Oct-24".

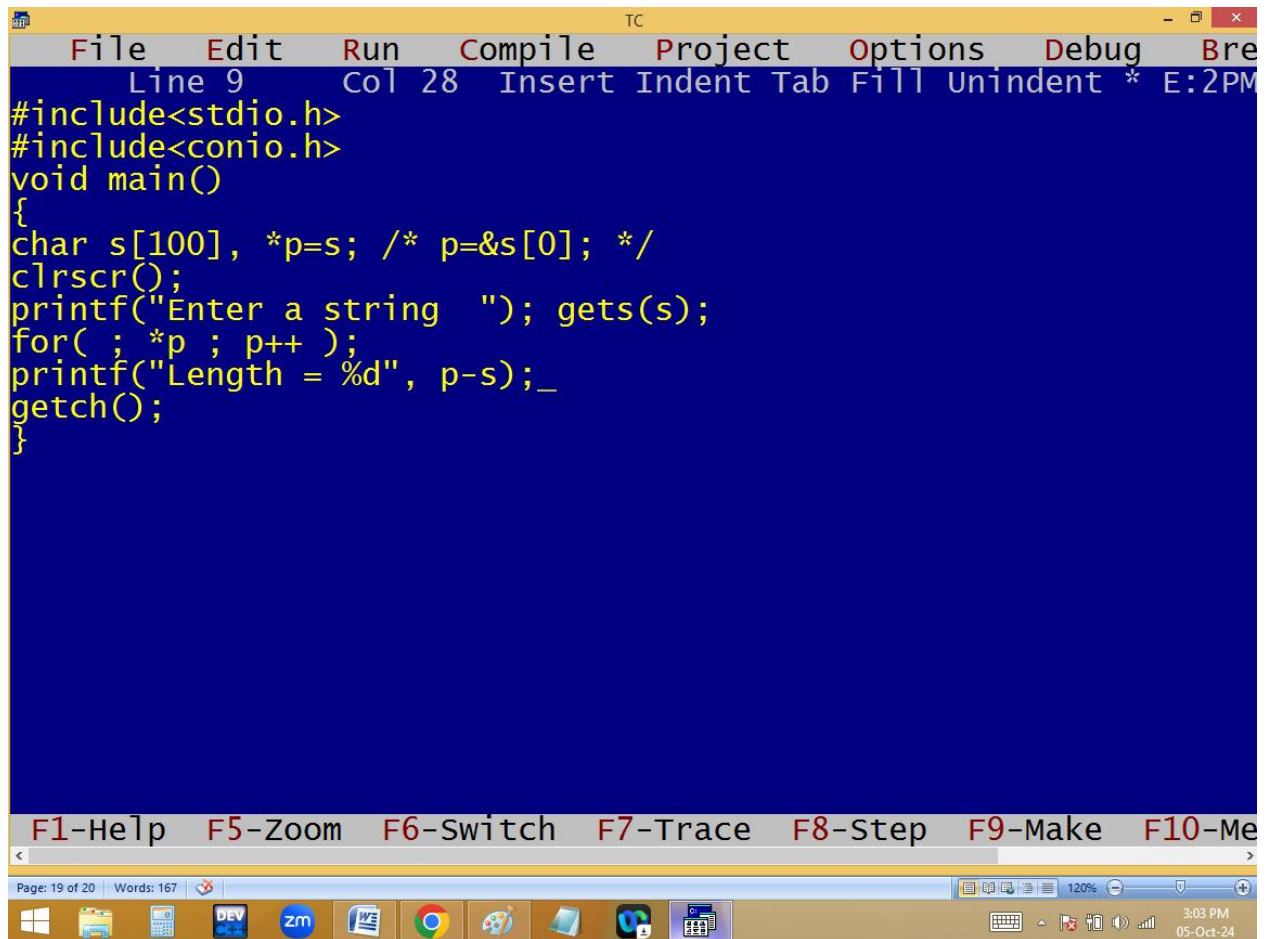
```
TC
Enter a string Bharathi
Bharathi
harathi
arathi
rathi
athi
thi
hi
i
_
```



```
for( ; *p!='\0'; p++) puts( p );
```

↳ 65500 to \0 = asha
65501 to \0 = sha
65502 to \0 = ha
65503 to \0 = a

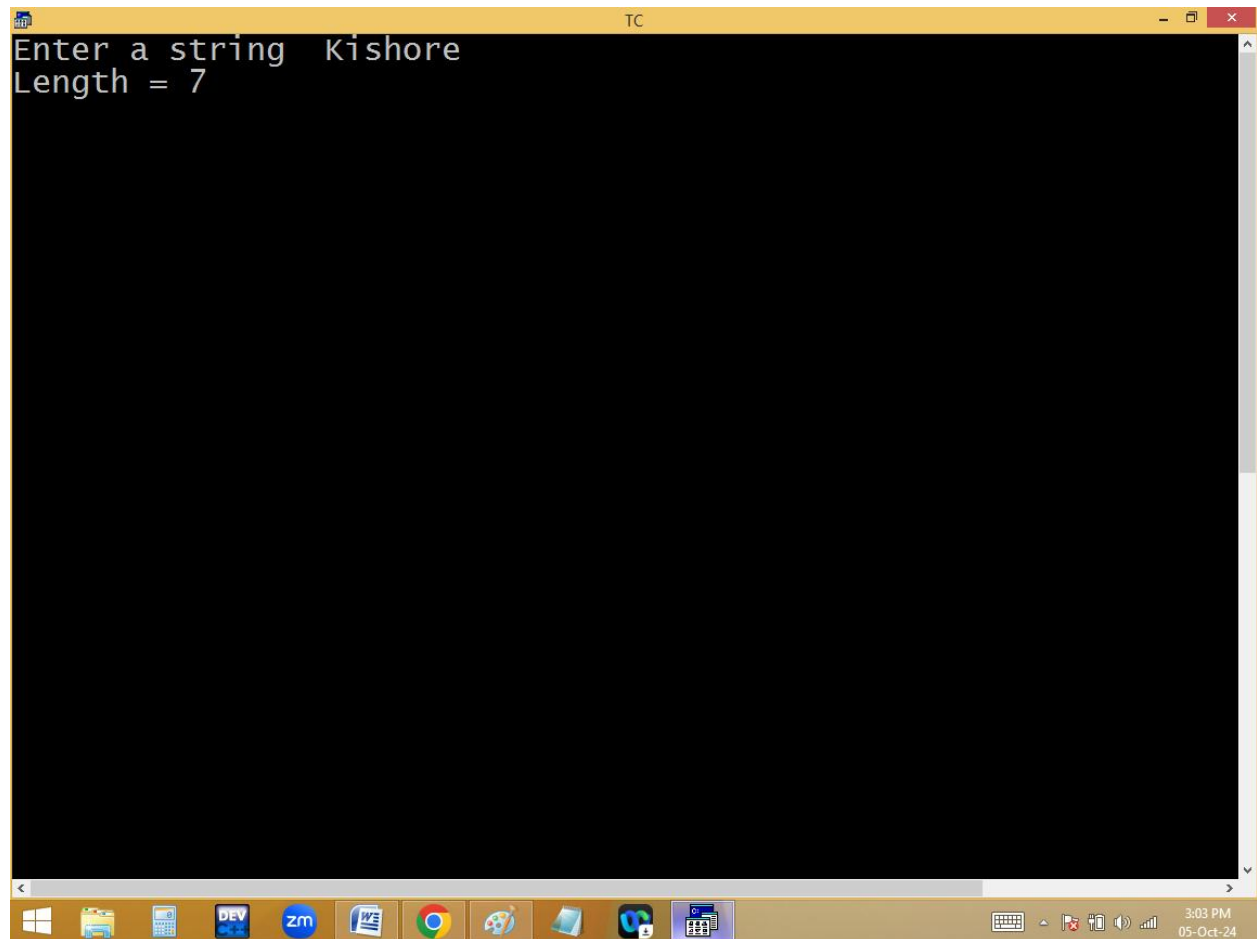
Find the string length using pointer only:

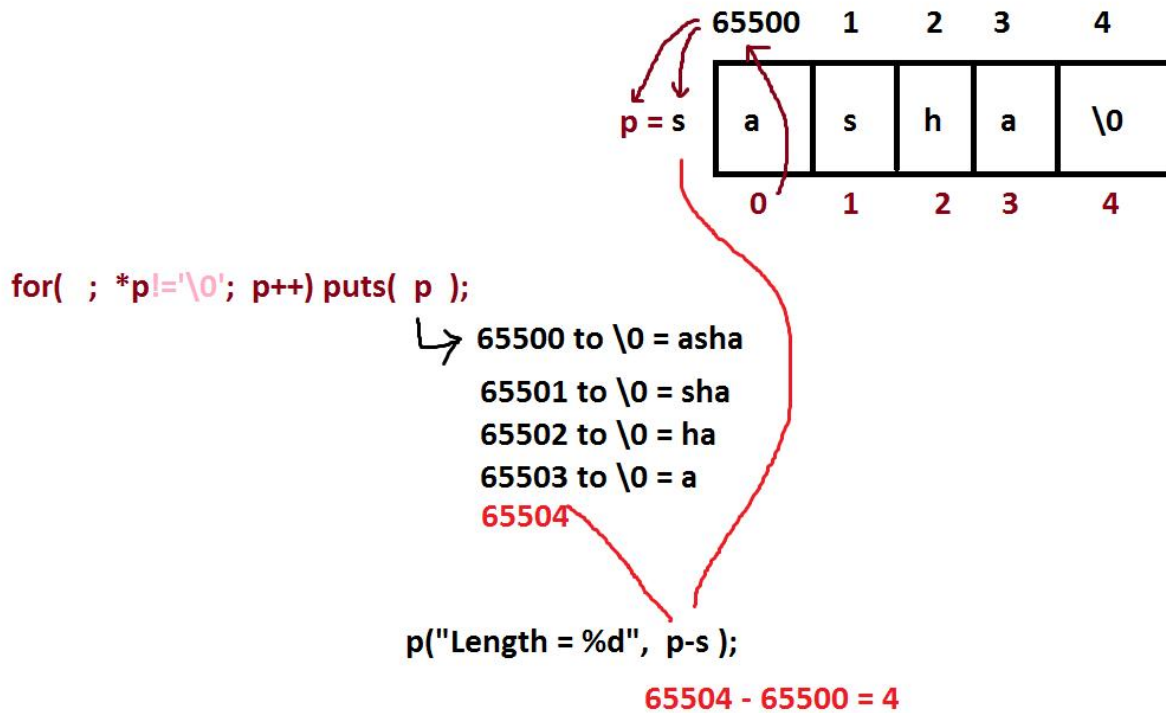


The image shows a screenshot of the Turbo C++ (TC) IDE. The main window has a blue background with yellow text for the code. The menu bar at the top includes File, Edit, Run, Compile, Project, Options, Debug, and Break. The status bar at the top shows 'Line 9', 'Col 28', and 'Insert Indent Tab Fill Unindent * E:2PM'. The code in the editor is as follows:

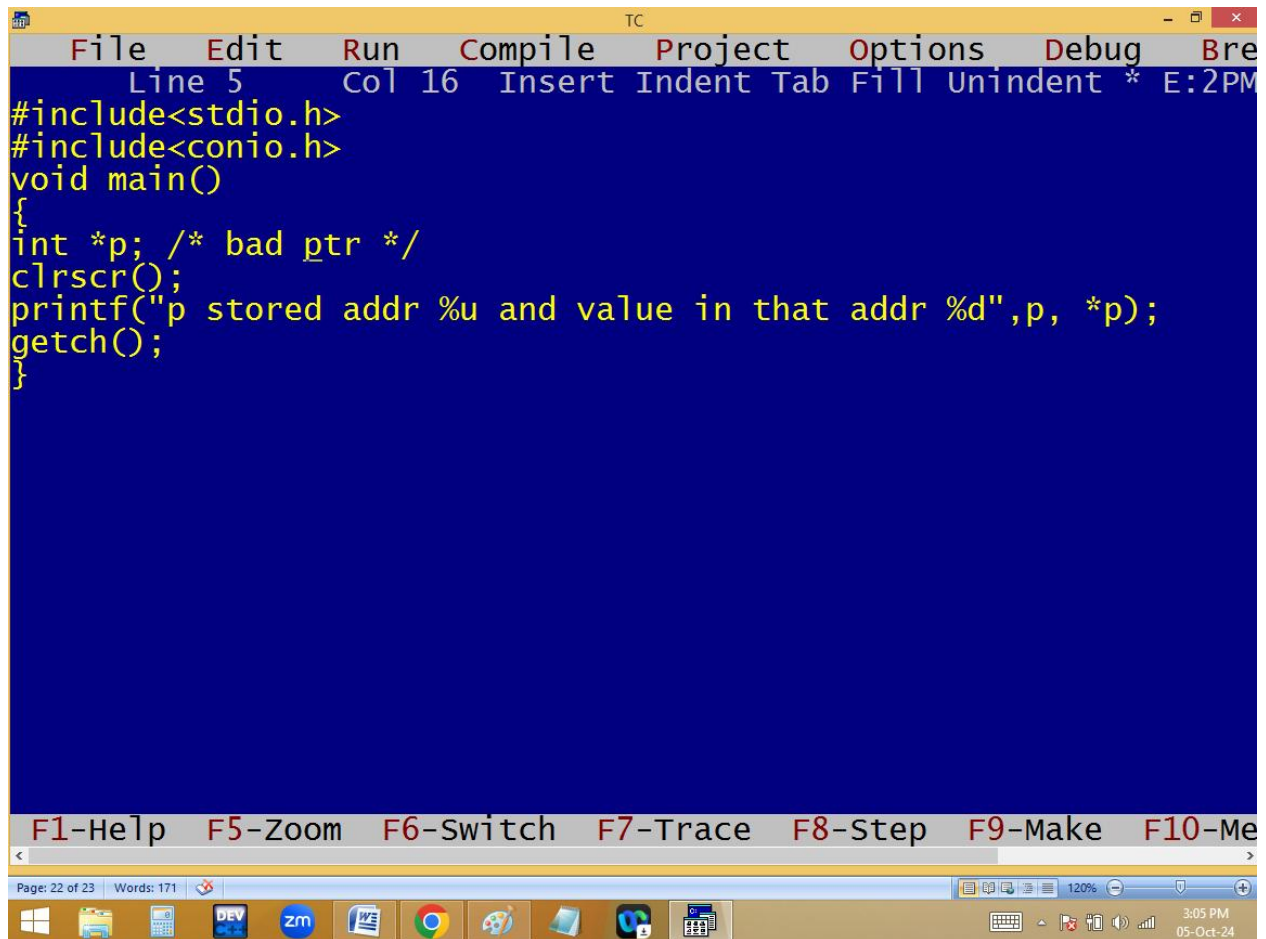
```
#include<stdio.h>
#include<conio.h>
void main()
{
char s[100], *p=s; /* p=&s[0]; */
clrscr();
printf("Enter a string "); gets(s);
for( ; *p ; p++ );
printf("Length = %d", p-s);_
getch();
}
```

Below the code editor, there is a toolbar with icons for various functions. At the bottom of the window, there is a status bar showing 'Page: 19 of 20' and 'Words: 167'. The Windows taskbar is visible at the very bottom, showing icons for Windows, File Explorer, Calculator, DEV C++, Zoom, Word, Chrome, and other applications. The system clock in the bottom right corner shows '3:03 PM' and '05-Oct-24'.





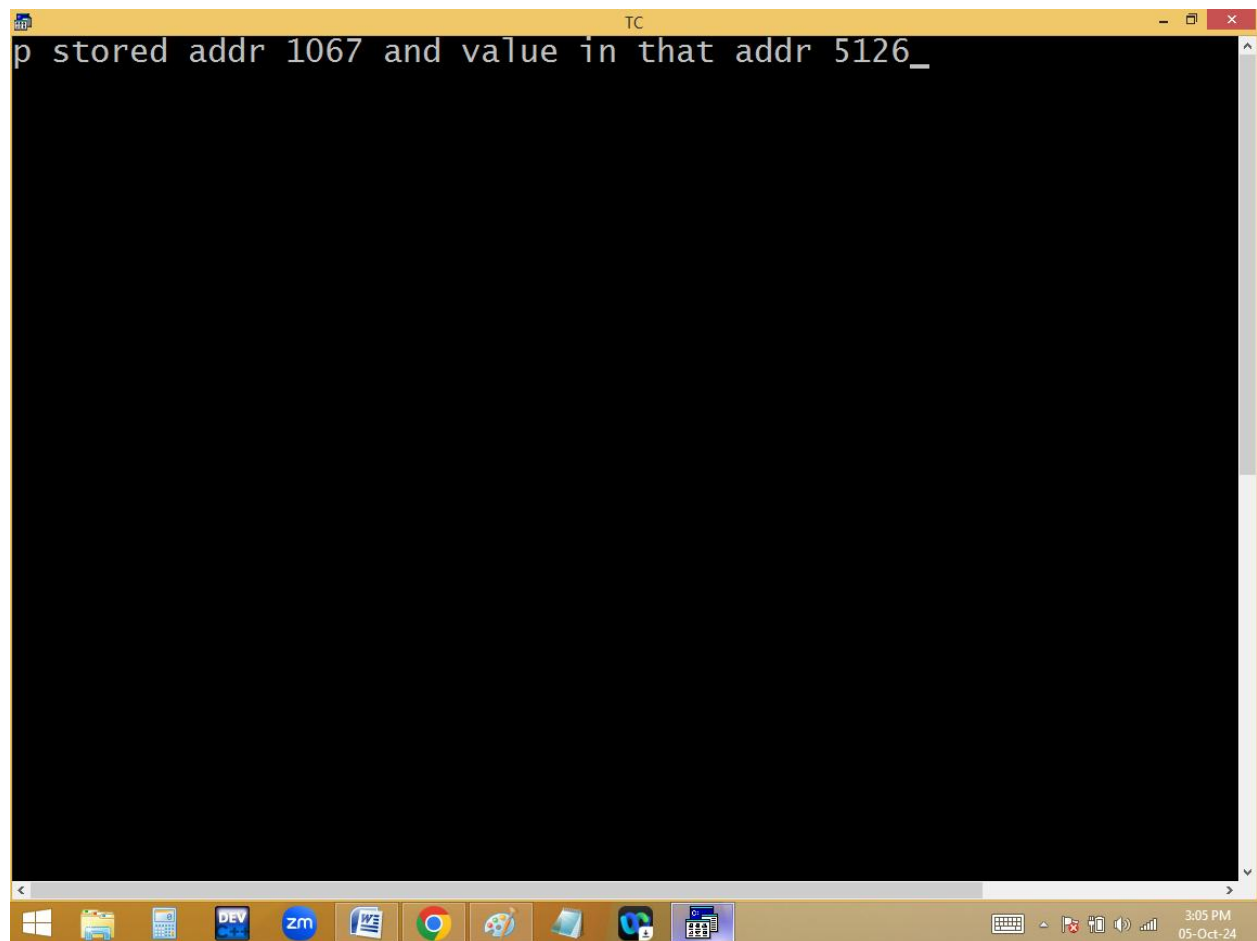
Bad / wild pointer: A pointer is declared but not initialized. In this situation the pointer is storing some unknown address and value. This kind of pointer is called bad / wild pointer.



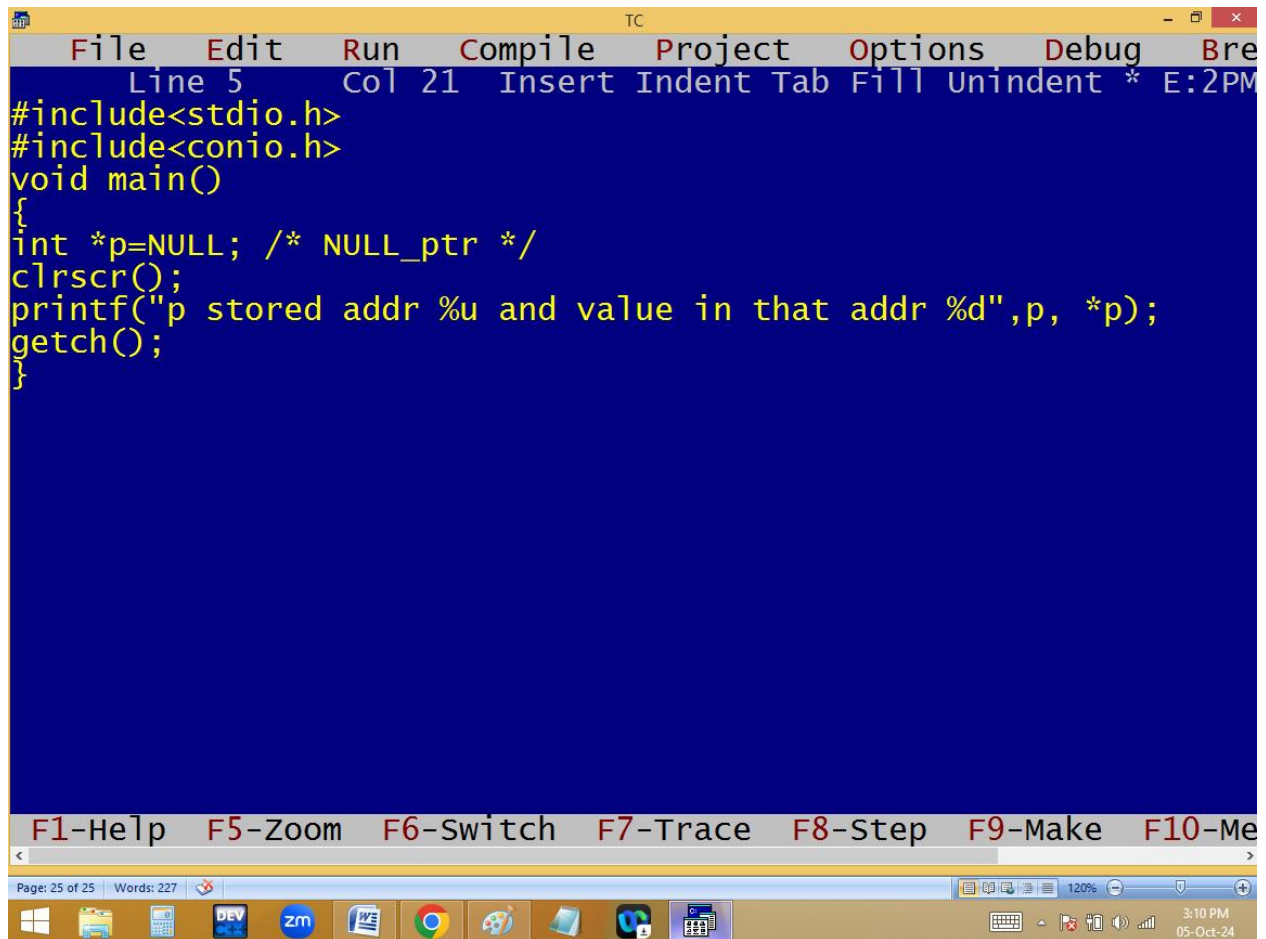
The image shows a screenshot of the Turbo C++ (TC) IDE. The main window has a dark blue background with yellow text. The menu bar at the top includes File, Edit, Run, Compile, Project, Options, Debug, and Break. The status bar at the top shows "Line 5", "Col 16", and "Insert Indent Tab Fill Unindent * E:2PM". The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int *p; /* bad ptr */
clrscr();
printf("p stored addr %u and value in that addr %d",p, *p);
getch();
}
```

Below the code, there is a toolbar with function key shortcuts: F1-Help, F5-Zoom, F6-Switch, F7-Trace, F8-Step, F9-Make, and F10-Me. At the bottom, there is a status bar showing "Page: 22 of 23" and "Words: 171". The Windows taskbar is visible at the very bottom, showing various application icons and the system clock indicating 3:05 PM on 05-Oct-24.



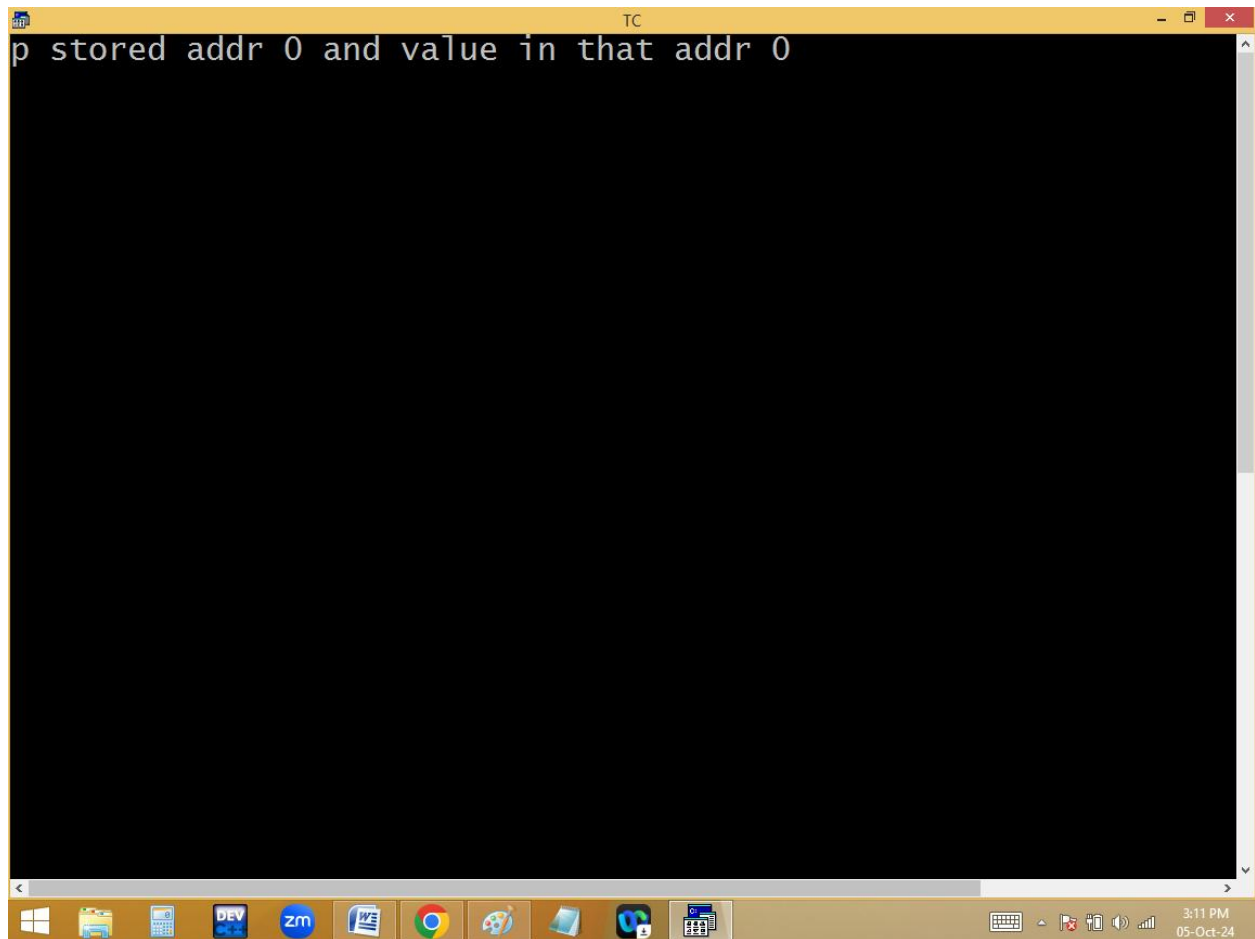
NULL pointer: When a pointer is initialized with 0 or NULL, it is a null pointer. To avoid bad and dangling pointers we are using NULL pointer.



The image shows a screenshot of the Turbo C++ (TC) IDE. The main window has a dark blue background with yellow text. The menu bar at the top includes File, Edit, Run, Compile, Project, Options, Debug, and Break. The status bar at the top shows 'Line 5', 'Col 21', and 'E:2PM'. The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int *p=NULL; /* NULL_ptr */
    clrscr();
    printf("p stored addr %u and value in that addr %d",p, *p);
    getch();
}
```

Below the code editor, there is a toolbar with icons for various functions. At the bottom of the IDE window, a status bar displays 'Page: 25 of 25' and 'Words: 227'. The Windows taskbar is visible at the very bottom, showing icons for Windows, File Explorer, Calculator, DEV C++, Zoom, Word, Chrome, and other applications. The system clock in the bottom right corner shows '3:10 PM' and '05-Oct-24'.



The image shows a screenshot of a Turbo C++ (TC) window. The window has a yellow title bar with the text "TC" and standard window controls. The main area is black, and the text "p stored addr 0 and value in that addr 0" is displayed in white. The Windows taskbar is visible at the bottom, showing various application icons and the system clock indicating 3:11 PM on 05-Oct-24.

```
p stored addr 0 and value in that addr 0
```

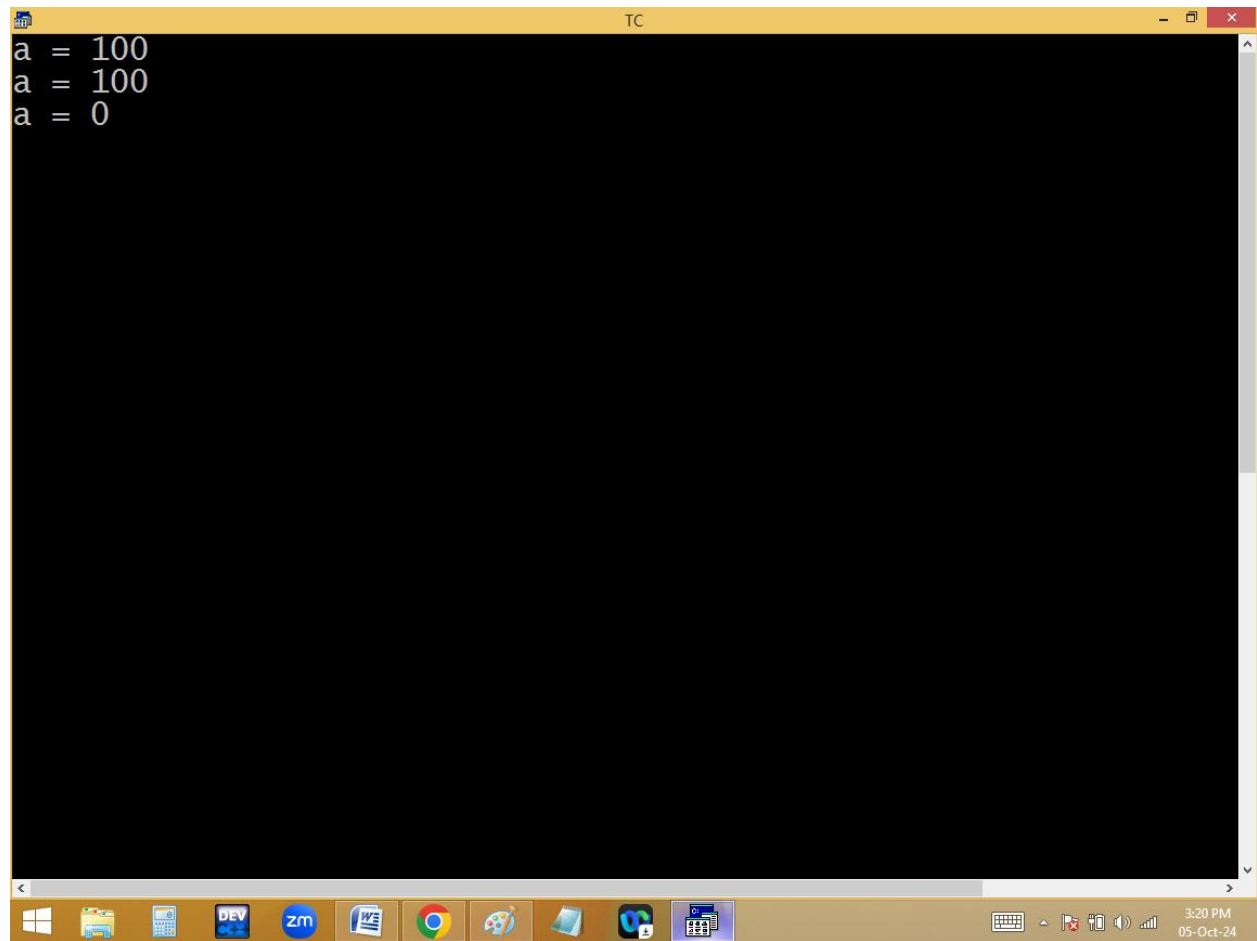
Dangling pointer: A pointer is declared and some variable address also assigned. After some time that variable deleted from memory. But still the pointer is storing the deleted variable address and showing the value. This kind of pointer is called dangling pointer and to avoid this initialize with NULL pointer.

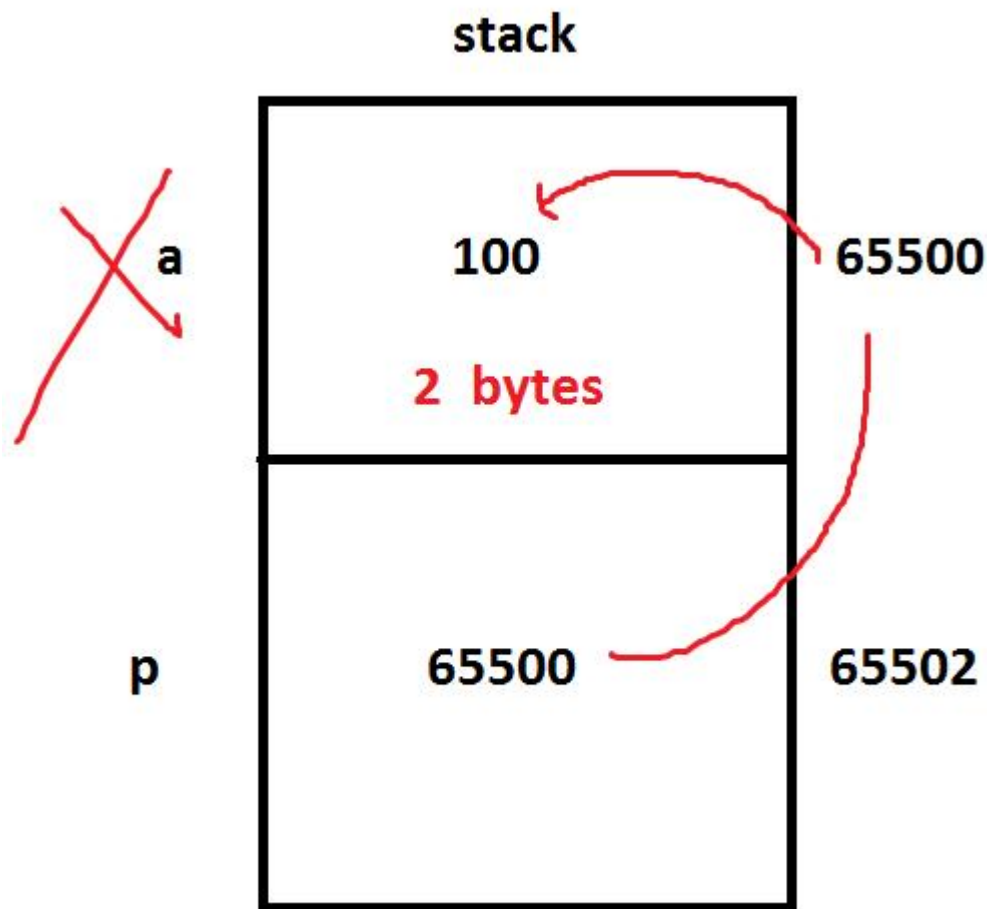
```
TC
File Edit Run Compile Project Options Debug Bre
Line 13 Col 4 Insert Indent Tab Fill Unindent * E:2PM
#include<stdio.h>
#include<conio.h>
void main()
{
int *p; /* bad ptr */
clrscr();
{
int a=100; /* local var */
p = &a;
printf("a = %d\n",*p);
} /* a deleted */
printf("a = %d\n", *p); /* dangling ptr */
p=0; /* NULL pointer */
printf("a = %d",*p);
getch();
}
```

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Me

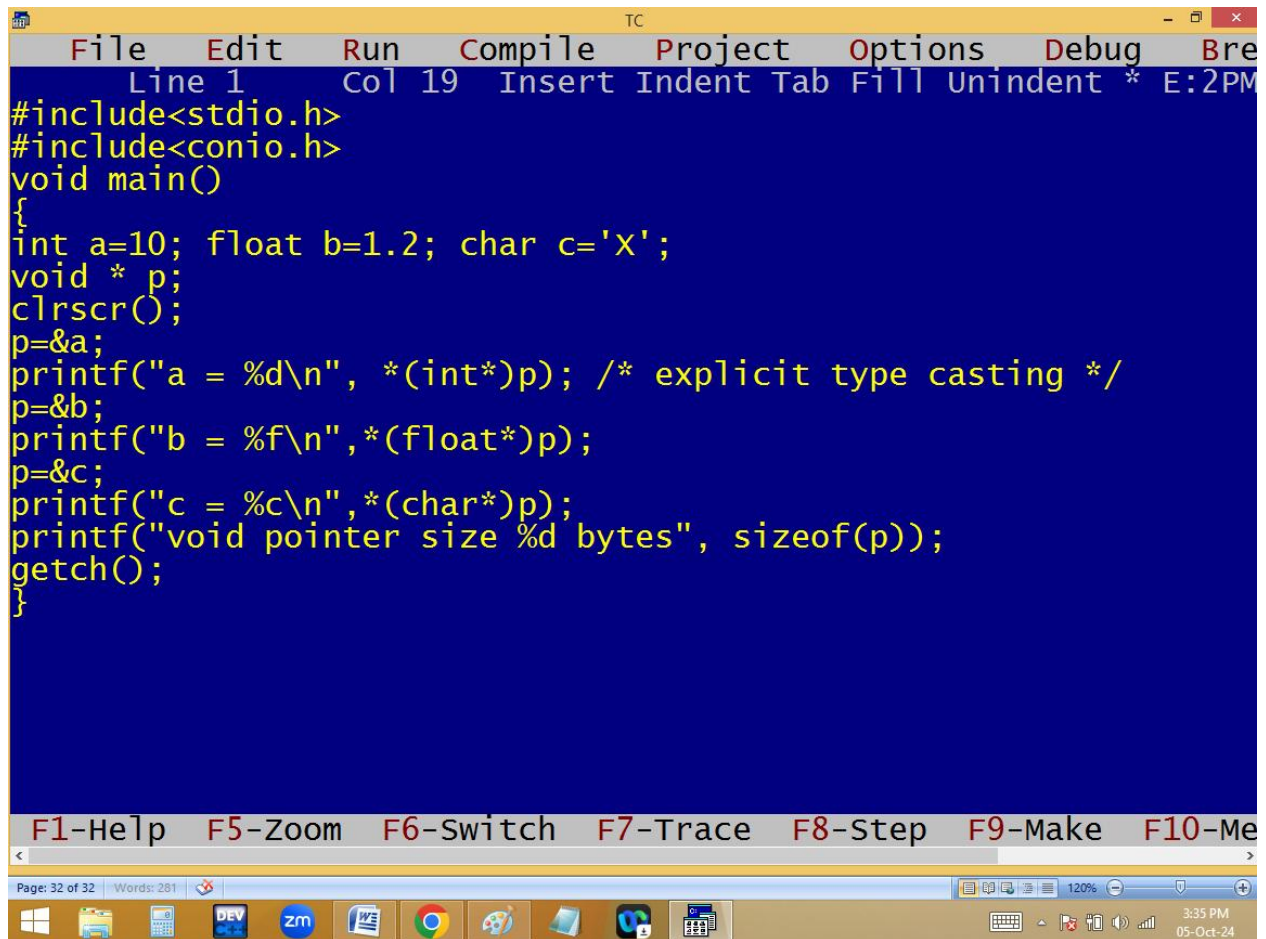
Page: 27 of 28 Words: 277 120% 3:20 PM 05-Oct-24

```
TC
a = 100
a = 100
a = 0
```





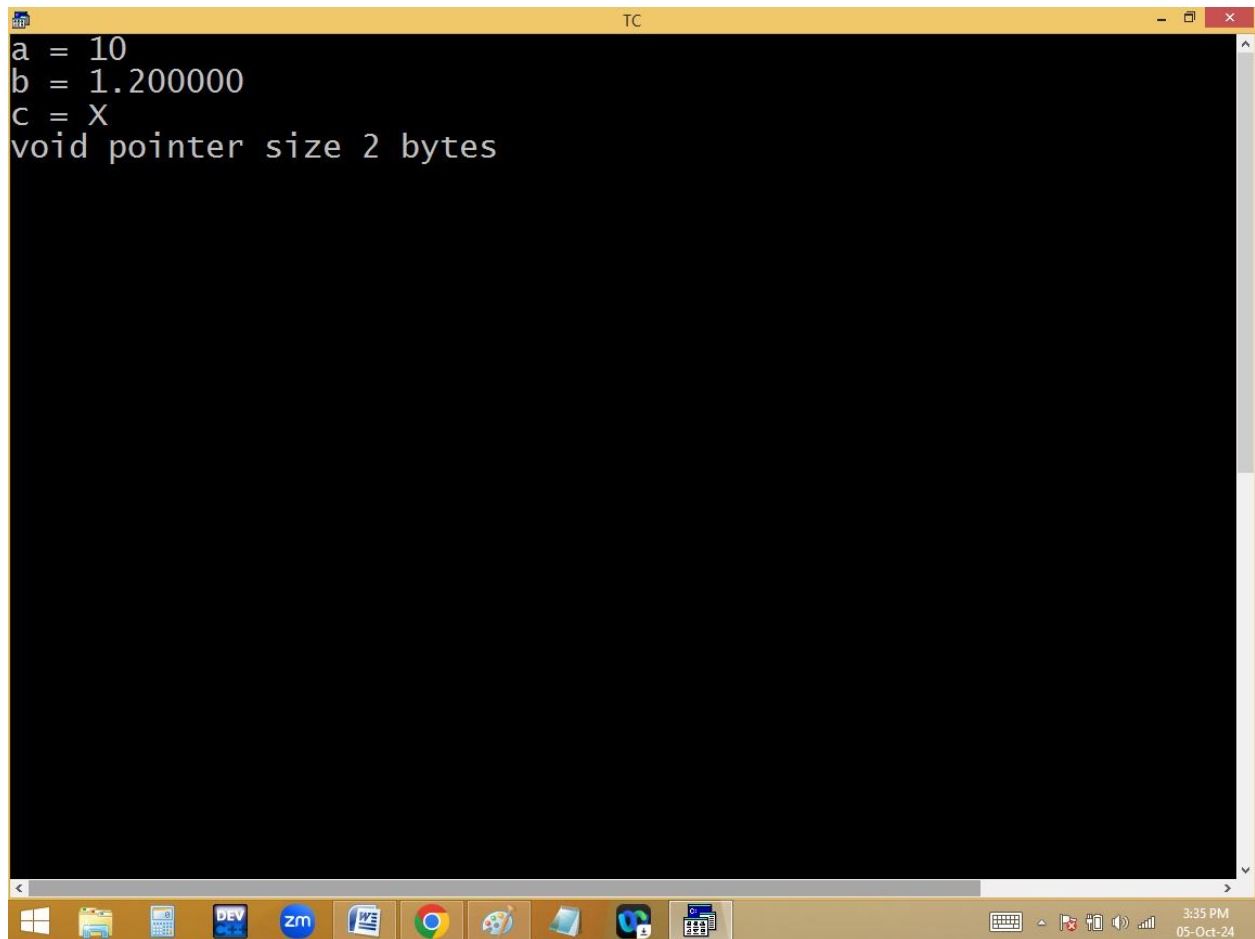
void / generic pointer: void pointer can store any type of address. It is used in dynamic memory allocation. Before going to use void pointer, explicit type casting should be conducted. It takes 2 bytes memory.



```
TC
File Edit Run Compile Project Options Debug Bre
Line 1 Col 19 Insert Indent Tab Fill Unindent * E:2PM
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10; float b=1.2; char c='X';
void * p;
clrscr();
p=&a;
printf("a = %d\n", *(int*)p); /* explicit type casting */
p=&b;
printf("b = %f\n",*(float*)p);
p=&c;
printf("c = %c\n",*(char*)p);
printf("void pointer size %d bytes", sizeof(p));
getch();
}
```

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Me

Page: 32 of 32 Words: 281 120% 3:35 PM 05-Oct-24

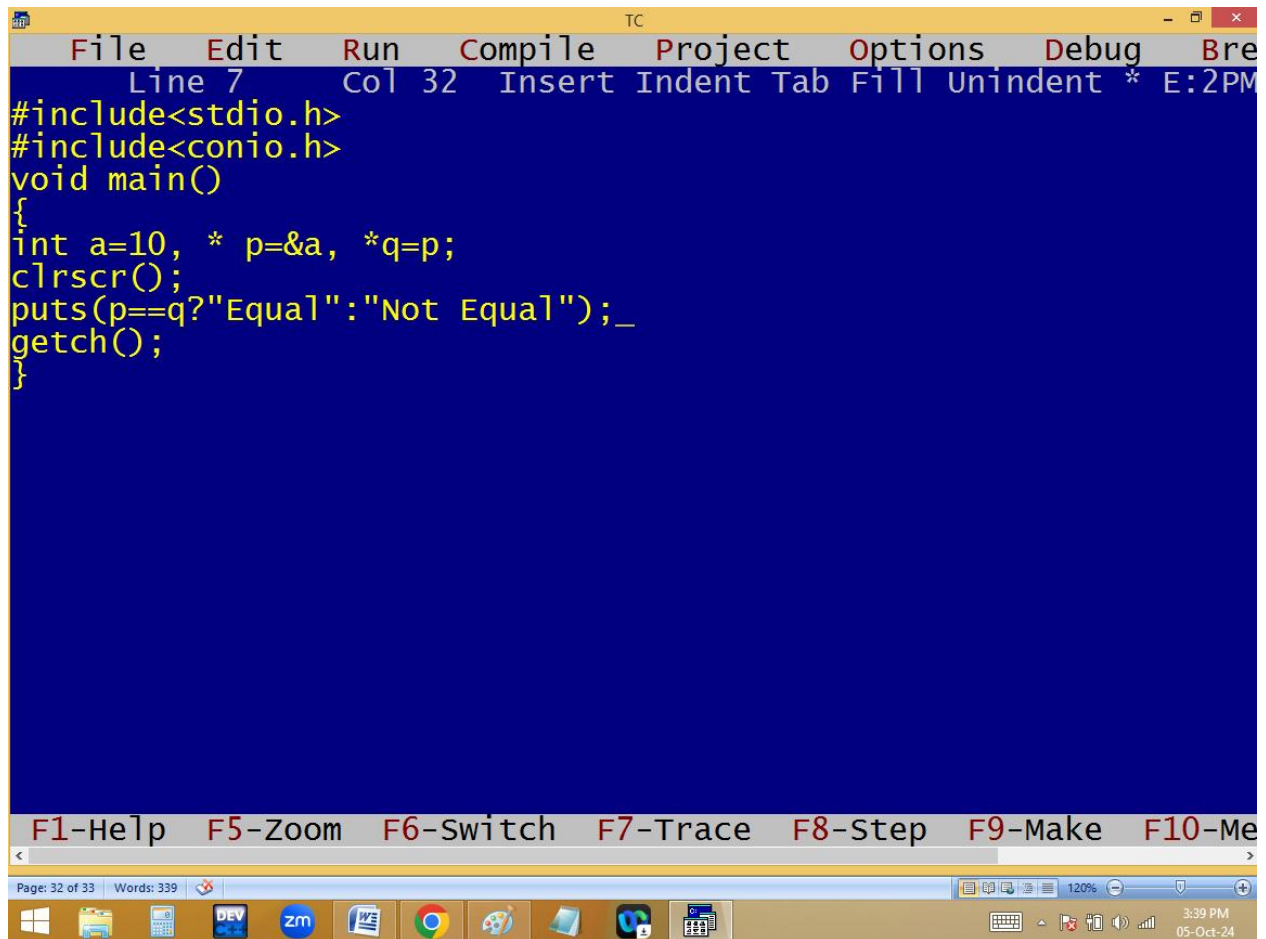


```
a = 10
b = 1.200000
c = X
void pointer size 2 bytes
```

The image shows a screenshot of a Turbo C++ (TC) compiler window. The window has a yellow title bar with the text "TC" and standard Windows window controls (minimize, maximize, close). The main area is black with white text. The text shows four lines of code: `a = 10`, `b = 1.200000`, `c = X`, and `void pointer size 2 bytes`. The bottom of the window shows a Windows taskbar with various icons including the Start button, File Explorer, Calculator, DEV, zm, Word, Chrome, Paint, and others. The system clock in the bottom right corner shows "3:35 PM" and "05-Oct-24".

Pointer arithmetic:

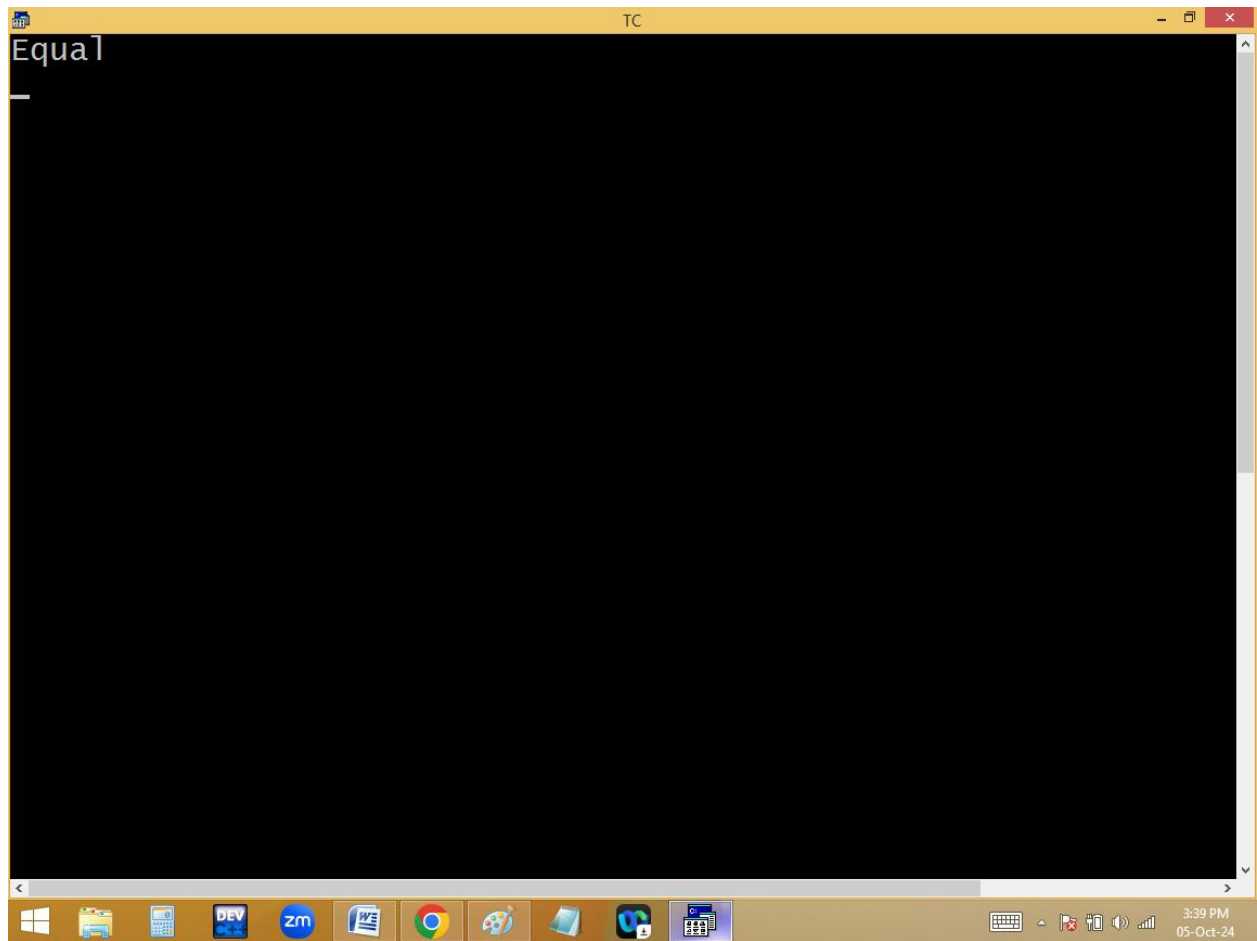
Like normal variables we can do +, -, ++, ==, copy, comparison on pointers also. But we can't do *, %, / on pointers.

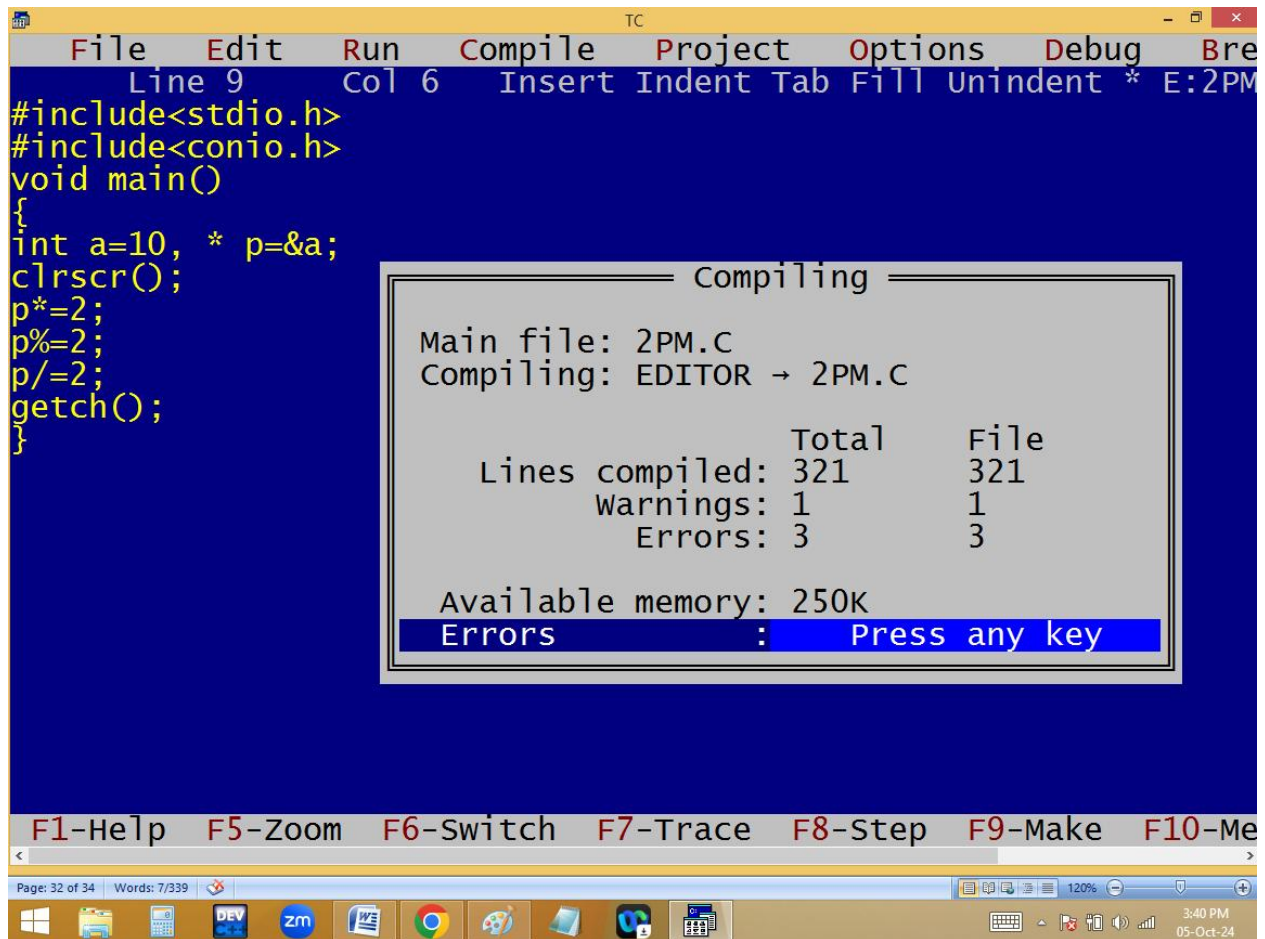


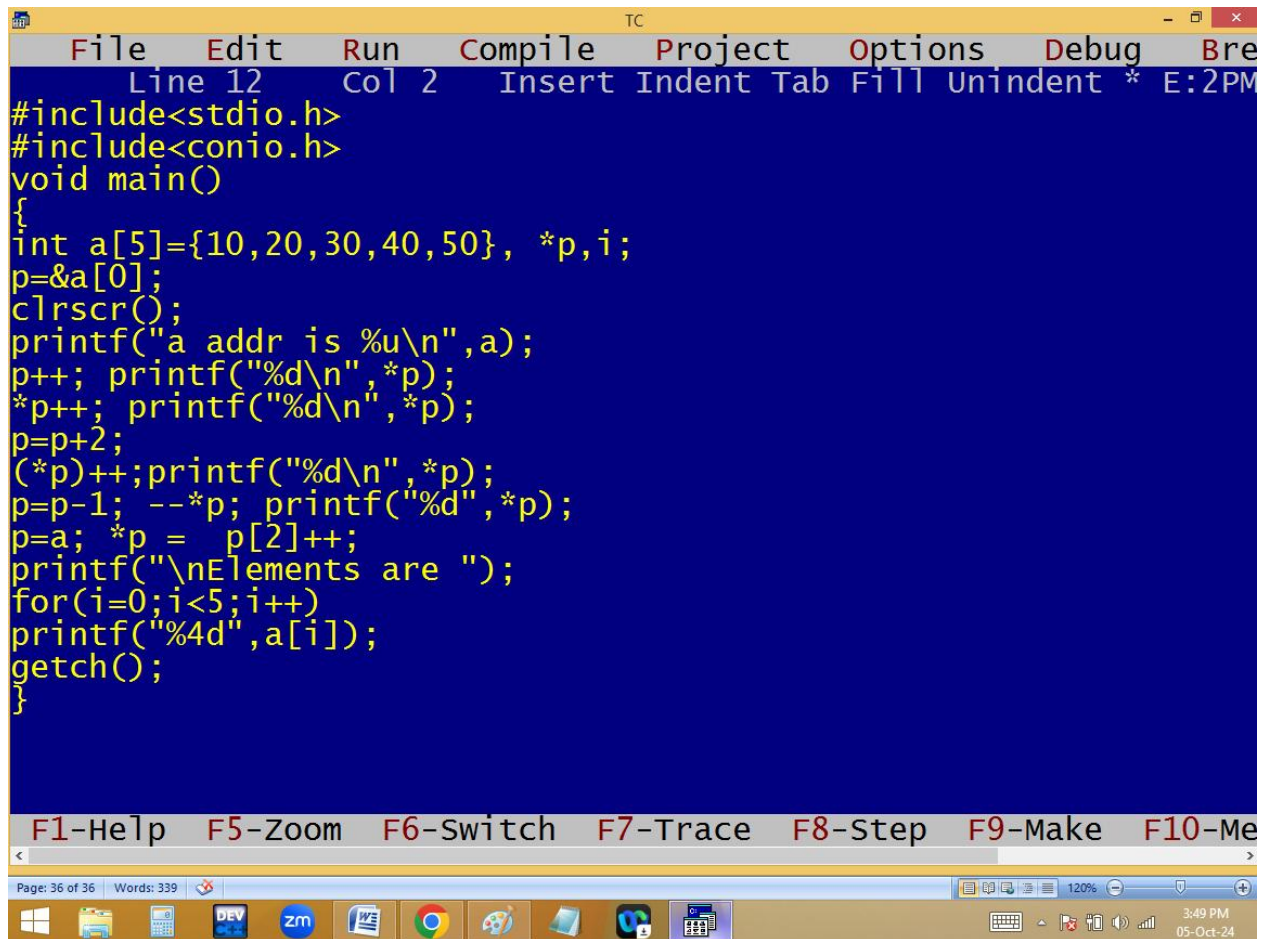
The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break. The status bar at the top shows "Line 7", "Col 32", and "Insert Indent Tab Fill Unindent * E:2PM". The code editor contains the following C program:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10, * p=&a, *q=p;
clrscr();
puts(p==q?"Equal":"Not Equal");_
getch();
}
```

Below the code editor, there is a toolbar with function key shortcuts: F1-Help, F5-Zoom, F6-Switch, F7-Trace, F8-Step, F9-Make, and F10-Me. The bottom status bar shows "Page: 32 of 33" and "Words: 339". The Windows taskbar at the bottom includes icons for Windows, File Explorer, Calculator, DEV C++, Zoom, Word, Chrome, Paint, and other applications. The system clock shows "3:39 PM" and "05-Oct-24".







```
TC
File Edit Run Compile Project Options Debug Bre
Line 12 Col 2 Insert Indent Tab Fill Unindent * E:2PM
#include<stdio.h>
#include<conio.h>
void main()
{
int a[5]={10,20,30,40,50}, *p,i;
p=&a[0];
clrscr();
printf("a addr is %u\n",a);
p++; printf("%d\n",*p);
*p++; printf("%d\n",*p);
p=p+2;
(*p)++;printf("%d\n",*p);
p=p-1; --*p; printf("%d",*p);
p=a; *p = p[2]++;
printf("\nElements are ");
for(i=0;i<5;i++)
printf("%4d",a[i]);
getch();
}
```

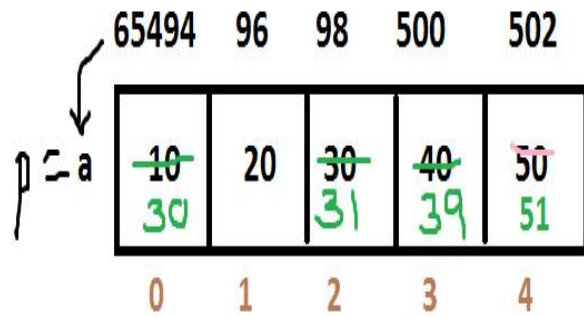
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Me

Page: 36 of 36 Words: 339 120% 3:49 PM 05-Oct-24

```
TC
a addr is 65494
20
30
51
39
Elements are 30 20 31 39 51_
```

$p++ \Rightarrow 65494 + 1 * 2 = 65496$
 $p(*p) \Rightarrow \text{value at } 65496 \Rightarrow 20$

$*p++ \Rightarrow 65496 + 1 * 2 = 65498$
 $p(*p) \Rightarrow \text{value at } 65498 \Rightarrow 30$



$p = p + 2 \Rightarrow 65498 + 2 * 2 = 65502$
 $(*p)++ \Rightarrow \text{value at } 65502++ \Rightarrow 50++ \Rightarrow 51$
 $p(*p) \Rightarrow \text{value at } 65502 \Rightarrow 51$

$p = p - 1 \Rightarrow 65502 - 1 * 2 = 65500$
 $--*p \Rightarrow \text{value at } 65500-- \Rightarrow 40-- \Rightarrow 39$
 $p(*p); \Rightarrow \text{value at } 65500 \Rightarrow 39$

$p = a \Rightarrow p = 65494$
 $*p = p[2]++; *p = p[2];$
 $65494 = 65494 + 2 * 2 = 65498 \Rightarrow 10 = 30$
 $p[2]++ \Rightarrow 30++ \Rightarrow 31$